Biochemistry and Chemical Biology

Reviewed Preprint
v2 • October 2, 2024
Revised by authors

Reviewed Preprint
v1 • September 11, 2023

Minimal twister sister (TS)-like self-cleaving ribozymes in the human genome revealed by deep mutational scanning

Zhe Zhang, Xu Hong, Peng Xiong, Junfeng Wang, Yaoqi Zhou, Jian Zhan

institute for Systems and Physical Biology, Shenzhen Bay Laboratory, Shenzhen, China • University of Science and Technology of China, Hefei, China • Institute for Glycomics, Griffith University, Southport, Australia • Suzhou Institute for Advanced Research, University of Science and Technology of China, Suzhou, China • High Magnetic Field Laboratory, Key Laboratory of High Magnetic Field and Ion Beam Physical Biology, Hefei Institutes of Physical Science, Chinese Academy of Sciences, Hefei, China • Institute of Physical Science and Information Technology, Anhui University, Hefei, China • School of Information and Communication Technology, Griffith University, Southport, Australia • Ribopeutic Inc, Guangzhou International Bio Island, Guangzhou, China

Abstract

Despite their importance in a wide range of living organisms, self-cleaving ribozymes in the human genome are few and poorly studied. Here, we performed deep mutational scanning and covariance analysis of two previously proposed self-cleaving ribozymes (LINE-1 and OR4K15). We found that the regions essential for ribozyme activities are made of two short segments, with a total of 35 and 31 nucleotides only. The discovery makes them the simplest known self-cleaving ribozymes. Moreover, the essential regions are circular permutated with two nearly identical catalytic internal loops, supported by two stems of different lengths. These two self-cleaving ribozymes, which shape like lanterns, are similar to the catalytic regions of the twister sister ribozymes in terms of sequence and secondary structure. However, the nucleotides at the cleavage site have shown that mutational effects on the two twister sisterlike (TS-like) ribozymes are different from the twister sister ribozyme. The discovery of TS-like ribozymes reveals a ribozyme class with the simplest and, perhaps, the most primitive structure needed for self-cleavage.

eLife Assessment

This **important** study uncovers a surprising link between two self-cleaving RNAs that belong to the same structural family. The evidence supporting the main conclusions is **convincing** and based on extensive biochemical and bioinformatic analysis. This research will be of broad interest to RNA molecular biologists and biochemists.

https://doi.org/10.7554/eLife.90254.2.sa2

Introduction

Ribozymes (catalytic RNAs) are thought to have dominated early life forms (1) until they were gradually replaced by more effective and stable proteins. One of the few that remain active is self-cleaving ribozymes found in a wide range of living organisms (2), involved in rolling circle replication of RNA genomes (3–5), biogenesis of mRNA and circRNA, gene regulation (6–9), and co-transcriptional scission of retrotransposons (10, 11).

So far, there are a few self-cleaving ribozymes found in the human genome. They include two hammerhead ribozymes (7, 12), a HDV-like cytoplasmic polyadenylation element-binding protein 3 (CPEB3) ribozyme within the transcript of a single-copy gene (8), and self-cleaving ribozymes associated with olfactory receptor OR4K15 (olfactory receptor family 4 subfamily K member 15), insulin-like growth factor 1 receptor (IGF1R), and a LINE-1 (Long Interspersed Nuclear Element-1) retrotransposon (8). All these ribozymes were discovered by using selection-based biochemical experiments (8). More recently, the hovlinc ribozyme (13) was identified by genome-wide screening against RNAs with the specific 5'-hydroxyl termini resulted from self-cleavage, followed by a combination of RNA structure prediction as well as deletion analysis and mutation-based validation for determining its secondary structure. It has been shown that single nucleotide polymorphism (SNP) in CPEB3 ribozyme was associated with an enhanced self-cleavage activity along with a poorer episodic memory (14). Inhibition of the highly conserved CPEB3 ribozyme could strengthen hippocampal-dependent long-term memory (15, 16). However, little is known about the other human self-cleaving ribozymes.

Previous studies on self-cleaving ribozymes found in retrotransposons suggested their importance in the overall size, structure, and function of different genomes. Most selfcleaving ribozymes found in retrotransposons belong to two widespread ribozyme families: HDV-like ribozymes in R2 elements (10) and LITc retrotransposon (11) and hammerhead ribozymes (HHR) in short interspersed nuclear elements (SINEs) of Schistosomes (17), Penelope-like elements (PLEs) (18, 19) and retrozymes (20). Unlike the wide occurrence of hammerhead ribozymes in different types of retrotransposons, HDV-like ribozymes usually locate in the 5' UTR region of the retrotransposons only, and were found in different species including *Drosophila melanogaster* (21, 22), *Bombyx mori* (23), arthropods, nematodes, birds and tunicates (24, 25). These ribozymes are important for co-transcriptional processing of the full-length transcript and translation of the downstream open reading frame (ORF). This suggests that the LINE-1 ribozyme found in the human genome may also play an active role in co-transcriptional processing during evolution.

Typically, self-cleaving ribozymes are in noncoding regions of the genome. For example, the CPEB3 ribozyme is located inside the intron of the CPEB3 gene (8), hammerhead ribozyme found in the 3' UTR of Clec2 genes (9), hammerhead and HDV motifs associated with retrotransposable elements (20, 24). However, the OR4K15 ribozyme is located at the reverse strand of the coding region of the olfactory receptor gene OR4K15, suggesting a potentially novel functional mechanism for this ribozyme.

Previously, we performed deep mutational scanning of a self-cleaving ribozyme (CPEB3) by using error-prone PCR to generate mutants in large scale. The relative activities of these mutants can be obtained by counting the cleaved and uncleaved sequences of each mutant from high-throughput sequencing data. These activities can be utilized to analyze mutational covariation and infer accurate base-pairing structures by developing a computational tool called CODA (covariation-induced deviation of activity) (26). Here, using the same technique, we located the regions essential for the catalytic activities (the functional regions) of LINE-1 and OR4K15 ribozymes in their original sequences, which were generated from selection of randomly fragmented human genomic DNA (8). We showed that these two ribozymes shared essentially the same base-pairing structural elements but with a circular permutation. The structural elements, which shaped like the Chinese lantern, are homologous to the catalytic cores of the more complex twister sister self-cleaving ribozymes in terms of sequence and secondary structure, suggesting a more primitive origin for twister sister ribozymes. However, the homology model of twister sister-like (TS-like) ribozyme generated from the twister sister ribozyme may not be the true structure for its function due to the lack of a stem loop for its stabilization and two mismatches from twister sister in the internal loops as well as different responses to mutations.

Results

Deep mutational scanning of LINE-1 and OR4K15 ribozymes.

Our previous work indicates that deep mutational scanning can lead to co-variation signals for highly accurate inference of base-pairing structures (26). Here, the same technique was applied to the original LINE-1 ribozyme (146 nucleotides, LINE-l-ori) and OR4K15 ribozyme (140 nucleotides, OR4K15- ori). These original sequences were generated from a randomly fragmented human genomic DNA Selection based biochemical experiments (8). Thus, their functional and structural regions are unknown. It needs to be noted that there is no exact match for LINE-l-ori and OR4K15-ori in the human genome (**Figure 1A**). This should be related to mutations accumulated during the selection. In deep mutational scanning, high-throughput sequencing can directly measure the relative cleavage activity (RA) (See Methods) of each variant in the mutant libraries of these two ribozymes according to the ratio of cleaved to uncleaved sequence reads of the mutant, relative to the same ratio of the wild-type sequence. The relative activities of single mutations at each position indicate the sensitivity of cleavage activity to the mutations in that position. **Figure 1B** shows the average RA for a given sequence position for all LINE-l-ori mutants with single mutations at the position. Mutations in most positions in LINE-l-ori did not lead to large changes in RA. In other words, these sequence positions are unlikely to contribute to the specific structure required for the ribozyme to be functional. The activity of the LINE-l-ori ribozyme is only sensitive to the mutations in two short segments (54–71 and 83–99) where the RA can be reduced to nearly zero. Thus, the two terminal ends and the central region between bases 72 and 82 are not that important for LINE-1 self-cleaving activity. In **Figure 1C**, a similar distribution of RA was observed in OR4K15-ori. Only sequence positions inside the two short segments (70–84 and 101–116) are sensitive to mutations, which suggests that only these regions are required to form the functional RNA structure of the OR4K15 selfcleaving ribozyme. Thus, we have identified the contiguous functional regions of LINE-1 ribozymes (54–99, LINE-l-rbz) and OR4K15 ribozymes (70–116, OR4K15-rbz).

The deep mutation data was further employed to search for the base pairs inside these two ribozymes by using CODA analysis. CODA employed Support Vector Regression (SVR) to establish an independent-mutation model and a naive Bayes classifier to separate bases paired from unpaired (**26**). Moreover, incorporating Monte-Carlo simulated annealing with an energy model and a CODA scoring term (CODA+MC) could further improve the coverage of the regions under-sampled by deep mutations. **Figures 1D and 1E** highlight the base pairs within the region 54–99 of LINE-l-ori (LINE-l-rbz) and the region 70–116 of OR4K15-ori (OR4K15- rbz), respectively. The CODA result for LINE-l-ori shows four base pairs (A14U34, A16U32, G17C31, C18G30) in one possible stem region, and another two base pairs (A4U43, A6A41) in the second possible stem region. Due to the lower mutational coverage, the CODA result for OR4K15-ori shows fewer base pairs than LINE-l-ori. Only two base pairs (U1A47, U3A45) in one possible stem region, and one lone base pair (A13U34) have been observed. The low mutation coverage for OR4K15-ori was due to the mutational bias (**27**,**28**) of error-prone PCR (**Supplementary Figures S1**, **S2**, **S3** and **S4**).

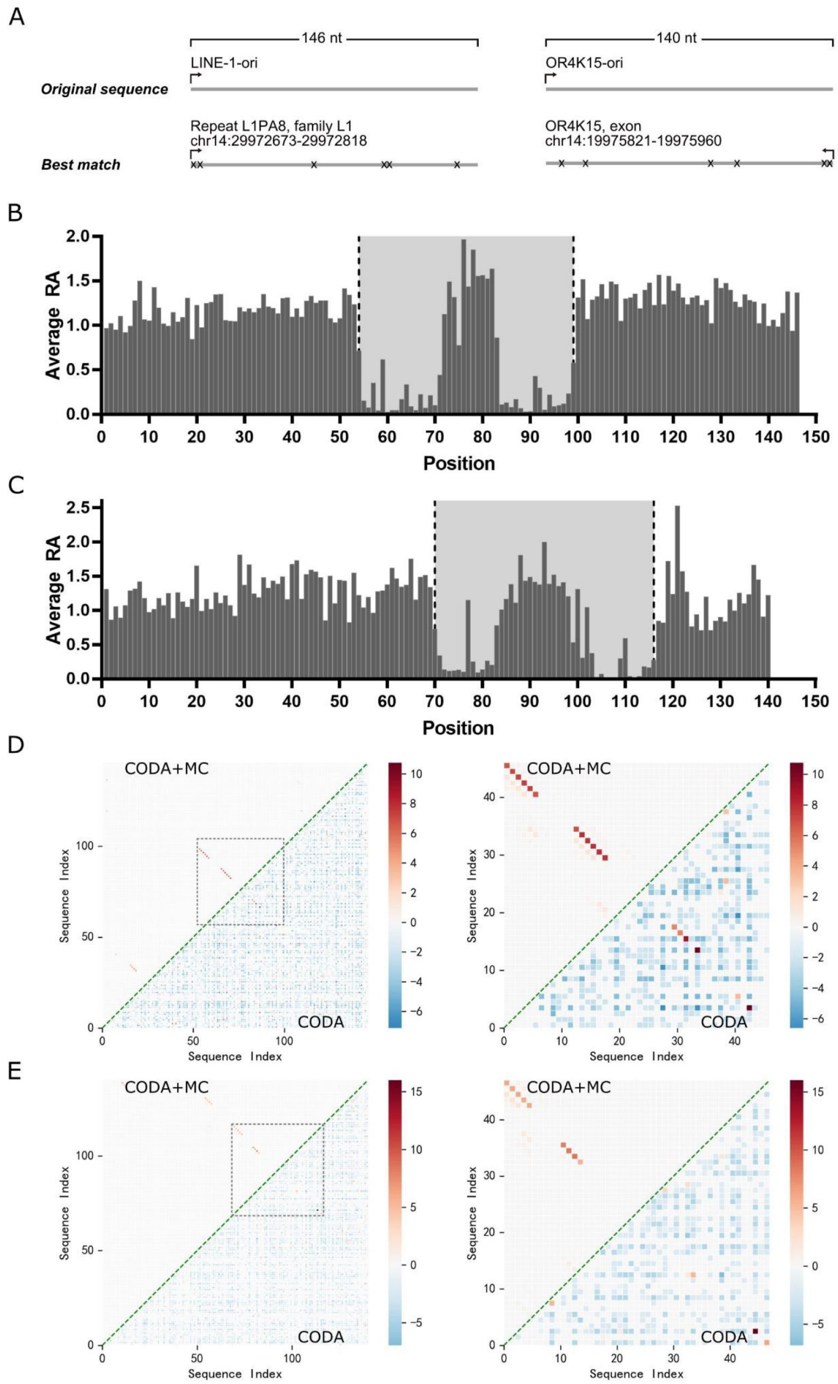


Figure 1.

Deep mutational scanning results of the LINE-1 and OR4K15 ribozymes.

(A) Genomic locations of the sequences with the highest similarity to the LINE-1 (left) and OR4K15 (right) ribozymes. (B) Average relative activity of mutations at each sequence position of the original LINE-1 ribozyme (LINE-l-ori). (C) Average relative activity of mutations at each sequence position of the original OR4K15 ribozyme (OR4K15-ori). (D) (Left) Base pairing maps inferred from LINE-l-ori deep mutation data by CODA (lower triangle) and by CODA in combination with Monte Carlo simulated annealing (CODA+MC) (upper triangle). (Right) Same as (Left) but for the contiguous functional region corresponds to 54–99 of LINE-l-ori (LINE-l-rbz). (E) (Left) Base pairing maps inferred from OR4K15-ori deep mutation data by CODA (lower triangle) and by CODA+MC (upper triangle). (Right) Same as (Left) but for the contiguous functional region which corresponds to 70–116 of OR4K15-ori (OR4K15-rbz).

A few but strong co-variation signals from CODA analysis can be expanded by MC simulated annealing as demonstrated previously (**26**). Several contiguous base-paired regions were detected (**Figures 1D and 1E**). LINE-l-rbz has two highly reliable stem regions with the length of 6 nt, including one non-Watson-Crick pair A6A41. For OR4K15-rbz, the functional region has two reliable stem regions with the lengths of 5 nt and 4 nt, respectively, including one non-Watson-Crick pair G14U33. Most of the base pairs discovered by MC simulated annealing are non-AU pairs, whose covariations are difficult to capture by error-prone PCR because of mutational biases (**27**, **28**). In addition, these two stems are consistent with the single mutation profiles. For example, two single mutations G14A and U33C showed high RA, 1.55 and 1.82, respectively, suggesting the high possibility of a G14U33 pair at these positions, because these two separate mutations change the wobble GU pair to the Watson-Crick AU and GC pairs, respectively, and lead to a more stable structure and higher activity. If G14U33 is true, a natural extension of the stem region is G15C32 for another canonical Watson-Crick pair. This pairing region was missed by the CODA+MC result. There are some other base pairs that appeared in low probabilities after MC simulated annealing. They are considered as false positives because of low probabilities and lack of support from the deep mutational scanning results. The appearance of false positives is likely due to the imperfection of the experiment-based energy function employed in current MC simulated annealing.

Further validation of base pairing information by deep mutational scanning of LINE-1- rbz.

To further improve the signals, we employed the contiguous functional segment (54–99, LINE-l-rbz) for the second round of deep mutational scanning (**Supplementary Figure S5**). Performing the second round is necessary because the deep mutational scanning of the original LINE-1 ribozyme has a low 18.5% coverage of double mutations (**Supplementary Table S1**). This low coverage is not sufficient for accurate inference of the secondary structure in the full coverage. This second round employed a chemically synthesized doped library with a doping rate of 6%, rather than a mutant library generated from error-prone PCR biased toward the sequence positions with A/T nucleotide (**Supplementary Tables S1**, **S2** and **Figures S1**, **S2**, **S3** and **S4**). In addition, to amplify the signals of cleaved RNAs after *in vitro* transcription of the mutant library of LINE-l-rbz, we selectively capture cleaved RNAs by employing RtcB ligase and a 5'-desbiotin, 3'-phosphate modified linker because they only react with the 5'-hydroxyl termini that exist only in cleaved RNAs (**29**). The captured active mutants were further enriched by the streptavidin-based selection after ligation. The technology improvement leads to less biased mutations in terms of the sequence positions (**Supplementary Figures S1** and **S2**) and mutation types (**Supplementary Figures S3** and **S4**). More importantly, it achieves 99.3% and 99.9% coverage of single and double mutations, respectively, within the contiguous functional segment (**Supplementary Table S1**).

Consistent secondary structure from deep mutational scanning of LINE-l-ori and LINE-l-rbz.

We performed the CODA analysis (**26**) based on the relative activities of 45,925 and 72,875 mutation variants (no more than 3 mutations) obtained for the original sequence and functional region of the LINE-1 ribozyme, respectively. CODA detects base pairs by locating the pairs with large covariation-induced deviations of the activity of a double mutant from an independent single-mutation model. **Figure 2A** shows the distribution of relative activity (RA', measured in the second round of mutational scanning) (See Methods) of all single mutations for the LINE-l-rbz ribozyme, which is consistent with the distribution of RA in the functional region of LINE-l-ori ribozyme (**Figure 1A** and **Supplementary Figure S5**). Moderate or high relative activity values (RA' > 0.5) at the two outer ends (18C, 30G, C46) of the stem regions suggest that these nucleotides might not be necessary for the function of LINE-l-rbz, whereas RA' values at 1G were affected by the transcription efficiencies of different nucleotides downstream the T7 promoter.

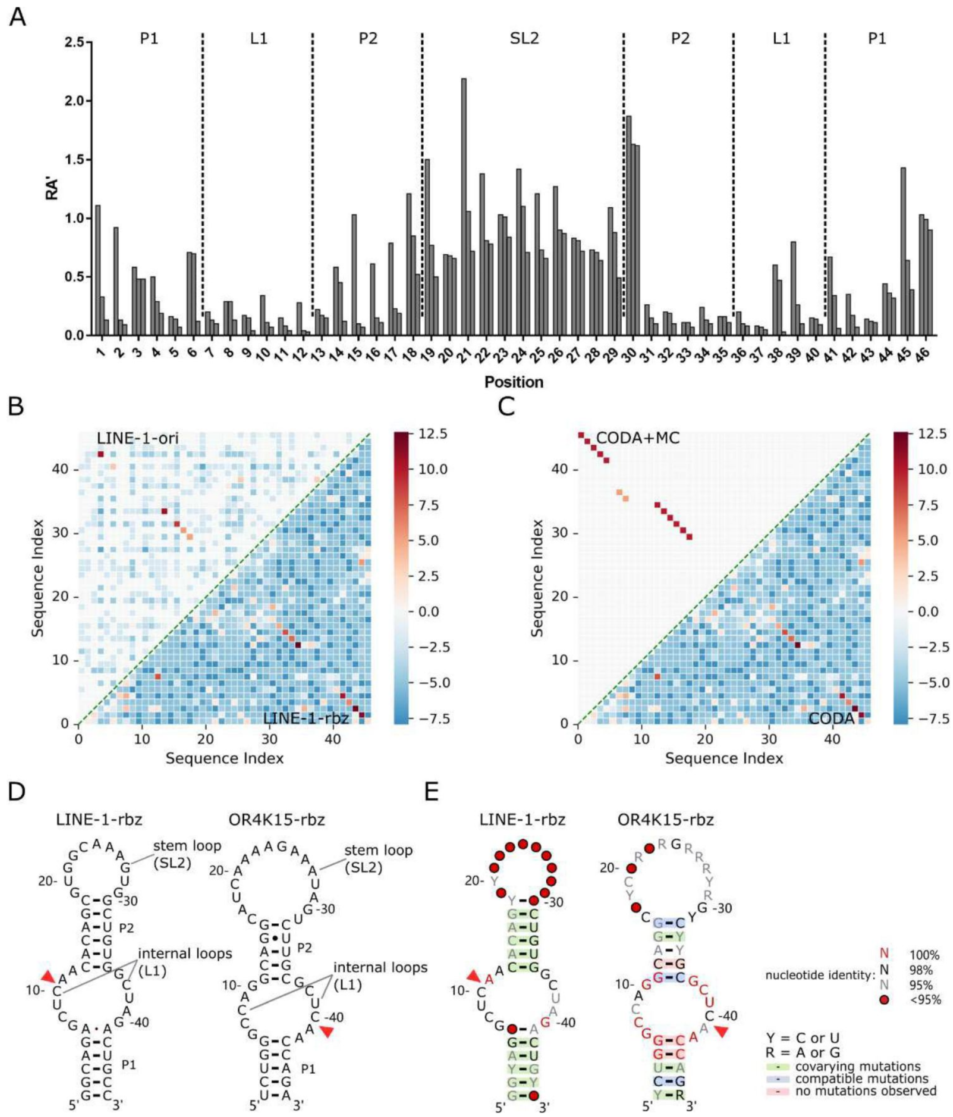


Figure 2.

Secondary structures of the functional regions of the LINE-1 and OR4K15 ribozymes.

(A) Relative activity of all mutations at each sequence position of the functional region of LINE-1 ribozyme (LINE-l-rbz). (B) Comparison between the base pairing maps inferred from CODA analysis of deep mutational scanning of the functional region of LINE-1 ribozyme (LINE-l-rbz) (lower triangle) and that from the corresponding region (54–99 nt) in the deep mutational scanning of the original LINE-1 ribozyme (LINE-l-ori) (upper triangle). (C) Comparison between the base pairing map inferred from LINE-l-rbz deep mutational data by CODA (lower triangle) and that after Monte-Carlo simulated annealing (CODA+MC, upper triangle). (D) Secondary structure model (left: LINE-l-rbz; right: OR4K15-rbz) inferred from the deep mutational scanning result. The red triangles indicate the cleavage sites of ribozymes. (E) Consensus sequence and secondary structure model (left: LINE-l-rbz; right: OR4K15-rbz) based on the alignment of functional mutants with RA’ ≥ 0.5. The red triangles indicate the ribozyme cleavage sites. Positions with conservation of 95, 98, and 100% were marked with gray, black and red nucleotides, respectively; positions in which nucleotide identity is less conserved are represented by circles. Green shading denotes predicted base pairs supported by covariation. R and Y denote purine and pyrimidine, respectively.

Figure 2B shows that data from both the original sequence and functional region of the ribozyme reveal the existence of two stem regions. The latter clearly shows two stems with lengths of 6 nt and 5 nt, respectively, due to nearly 100% coverage of single and double mutations for the functional region. Moreover, it suggests a non-canonical pair 6A41A at the end of the first stem, although the signal is weak. Such a weak signal is expected because the non-canonical pairs are typically less stable than standard base pairings (**30**). There are some additional weak signals in the LINE-l-rbz result. Most of these signals are for the base pairs at a few sequence distances apart (|i-j|<6, local base pairs). We found that some of them are caused by a high relative activity (RA') of the double mutant and are not likely to be true positives because the corresponding single mutations are not very disruptive (RA' > 0.5). The consistency between base pairs inferred from deep mutational scanning of the original sequences and that of the identified functional regions confirmed the correct identification of functional regions for LINE-1 ribozyme.

Probabilistic CODA results were further refined by Monte Carlo simulated annealing (CODA+MC). The resulting base pairs (**Figure 2C**, upper triangle) removed mostly local false positives. However, the noncanonical AA pair was removed as well, likely because the energy function does not account for noncanonical pairs. A new 18C30G pair added is a natural extension of the second stem. Two separate consecutive base pairs (7G37C, 8C36G) with relatively low signals were also added in the CODA+MC result. These two base pairs are likely false positives because they do not appear in all the models generated from Monte Carlo simulated annealing with a low probability of 0.52. Taking all the information together leads to a confident secondary structure for the LINE-l-rbz ribozyme, which contains two stem regions (Pl, P2), two internal loops, and a stem-loop (**Figure 2D** left).

The consistent result between LINE-l-rbz and LINE-l-ori suggested that reliable ribozyme structures could be inferred by deep mutational scanning. This allowed us to use OR4K15-ori to directly infer the final inferred secondary structure for the functional region of OR4K15. The secondary structures for these two ribozymes (**Figure 2D**) are surprisingly similar to each other. Both ribozymes have two stems (Pl, P2), two internal loops and a stem-loop region. Both stem-loop regions (SL2) are insensitive of ribozyme activity to mutations (**Figures 1B, 1C**). The internal loop regions (LI) of LINE-l-rbz and OR4K15-rbz are nearly identical except that C8 in OR4K15-rbz is replaced by U38 in LINE-l-rbz (**Figure 3A**).

Consensus sequence of LINE-l-rbz and OR4K15-rbz.

The consensus sequences (**Figure 2E**) for LINE-l-rbz and OR4K15-rbz were generated by R2R (**31**) based on the multiple sequence alignment of 1394 and 621 mutants with RA ≥ 0.5 from deep mutational scanning. **Figure 2E** overlays the consensus sequences onto the secondary structure models from the CODA+MC analysis. The self-cleavage of the two ribozymes occurs between a conserved CA dinucleotide which locates inside the longer part of the internal loops (LI) Unking P1 and P2. The internal loops are the regions with the highest conservation, which suggests their important roles in cleavage activity. Both stem regions (Pl, P2) are not very conserved because of covariations. For example, in **Figure 2E** (right), 1U47A can be replaced by 1C47G, and 14G33U will form 14C33G after double mutations. Covariation of all CG pairs was missing in OR4K15-rbz due to the mutational biases. The stem-loop regions show the lowest sequence identity, consistent with its insensitivity of catalytic activity to mutations in **Figures 1B and 1C**, providing additional support for the secondary structure models obtained here.

The secondary structure of OR4K15-rbz is a circular permutation of LINE-l-rbz selfcleaving ribozyme.

Removing the peripheral loop regions (**Figures 1B and 1C**) allows us to recognize that the secondary structure of OR4K15-rbz is a circular permutated version of LINE-l-rbz. As shown in **Figure 3A**, after a permutation, OR4K15-rbz has two conserved internal loops with one loop

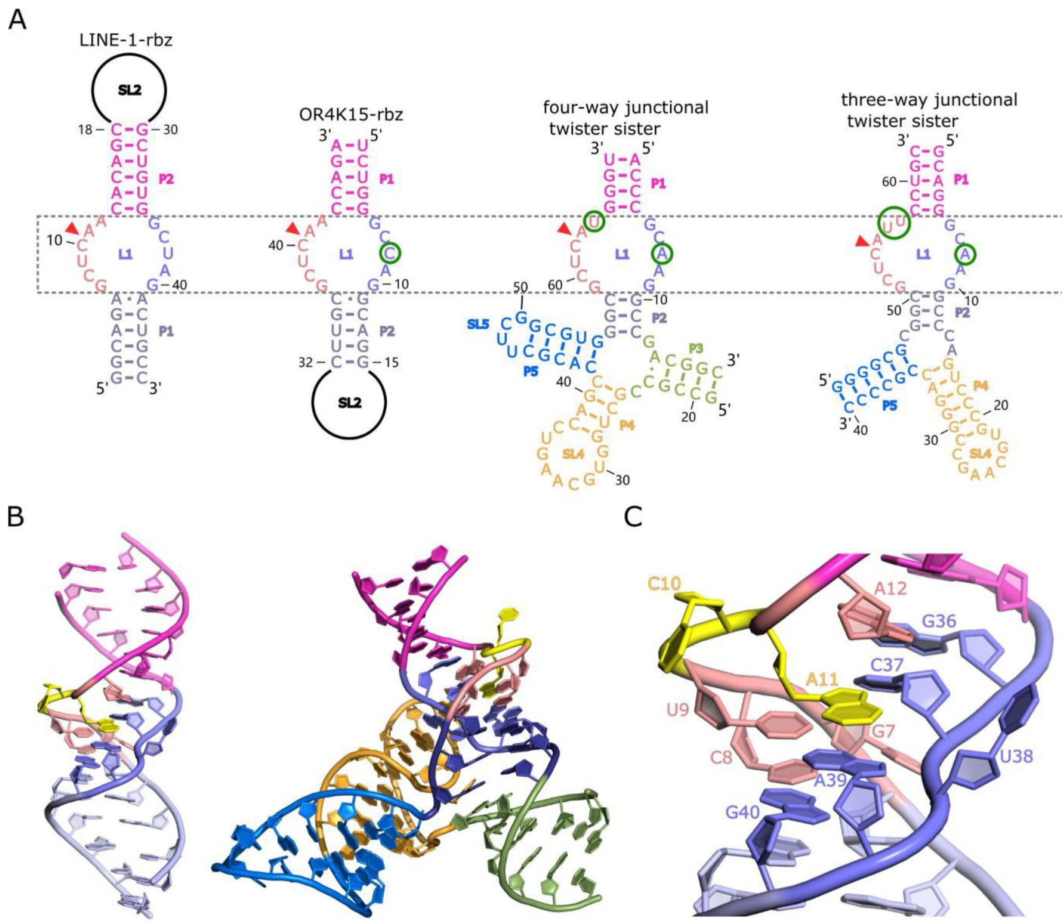


Figure 3.

Homology modelled structures of the bimolecular LINE-1-core and OR4K15-core.

(A) Comparison between the secondary structure of LINE-l-rbz, OR4K15-rbz, four-way (PDB ID: 5Y87) and three-way junctional twister-sister ribozyme (PDB ID: 5T5A) reveals strikingly similar internal loops surrounding the cleavage sites (red triangles) with few nucleotide differences at each side of the internal loops LL (B) A cartoon view of the homology modelled structure of LINE-1-core (left) and four-way junctional twister-sister ribozyme (PDB ID: 5Y87) (right). (C) A detailed view of the catalytic core LI of LINE-1-core in cartoon representation.

identical to LINE-l-rbz and the other loop differed from LINE-l-rbz by one base. The only mismatch U38C in LI has the RA' of 0.6, suggesting that the mismatch is not disruptive to the functional structure of the ribozyme. As the simplest ribozymes reported so far, it is important to know if these two ribozymes share the same motifs with the known ribozyme families.

Here we used pattern-based similarity search (RNAbob, *http://eddylab.org/software.html*) to search the structural patterns against the known ribozyme families in Rfam database (**Supplementary Figure S6**). The structural patterns of two ribozymes were derived from our deep mutational scanning results. We obtained three hits with identical motifs all from the twister sister ribozyme class (**Supplementary Table S3**). **Figure 3A** further shows the comparison with two previously published twister sister structures (**32**, **33**). The two internal loops of LINE-l-rbz with 5 and 6 nucleotides, respectively, differ from the catalytic internal loops of the four-way junctional twister sister ribozyme only by one base in each loop (**Figure 3A**) (**32**). Given the identical lantern-shaped regions of the LINE-l-rbz and OR4K15-rbz ribozyme, we named them twister sister-like (TS-like) ribozymes. Thus, it is possible to use the structure of twister sister ribozyme to build a homology model.

Homology modelling of the TS-like ribozymes.

To obtain the structure model of TS-like ribozymes, we used template-based homology modelling with the twister sister ribozyme structure as the template. **Figure 3B** shows the homology modelled structure of bimolecular LINE-1-core (internal loops plus stems) built from the more identical four-way junctional twister sister ribozyme (PDB ID: 5Y87). The structure of the four-way junctional twister sister ribozyme revealed the internal loops LI as the catalytic region involving a guanine-scissile phosphate interaction (G5-C62-A63), continuous stacking interactions, additional pairings and hydrated divalent Mg^{2+} ions. **Figure 3B, 3C** and **Supplementary Figure S7** show that the LINE-1-core model can be built as the twister sister ribozyme in the internal loops LI, with Al 1 at the cleavage site directed inwards. As shown in **Figures 3C** and **Supplementary Figure S8**, stem Pl of LINE-l-rbz is extended by the Watson-Crick U9A39 as a part of the G7(U9A39) base triple and Watson-Crick C8G40. Meanwhile, stem P2 of LINE-l-rbz is extended through trans non-canonical A12G36 and trans sugar edge-Hoogsteen Al 1C37. In addition to the triple base interaction in LINE-l-rbz, G7 is also H-bonded to both non-bridging O atoms of the phosphate connecting C37 and U38 (**Figure 4A**). This, along with the H-bond interaction between *proR* O atom of the phosphate and A39 N6, generates a stable square array of H-bonds.

We also modelled the structure of OR4K15-core in the same way as LINE-1-core. As shown in **Supplementary Figure S9A and S9B**, the model structure of LI in OR4K15-core is similar to LI in the LINE-1-core model, with stems P1 and P2 extended by additional base-pairing interactions in the tertiary structure of OR4K15-core. The base triple interaction of G37(A9U39) also exists in OR4K15-core, with the difference that G37 is H-bonded to A9 N6, rather than U39 (**Supplementary Figure S9C**). Thus, the square array of H-bonds could not be observed as there was no direct interaction between A9 and the phosphate connecting C7 and C8. This modelling result suggests that the bond between A9 (A39 in LINE-1-core) and the phosphate may not be important, consistent with the fact that the replacement of AU pair by a non-canonical UU pair did not disrupt cleavage activity (RA' =0.8 for A39U in LINE-l-rbz).

Like the two twister sister ribozyme structures (**32**, **33**), the nucleotide U38 in the LINE-1- core model (or C8 in the OR4K15-core model) is extruded from the helix (**Figure 3C**, **Supplementary Figure S9A**). The corresponding nucleotides, A8 in three-way junctional twister sister ribozyme (**33**) and A7 in four-way junctional twister sister ribozyme (**32**) are involved in the tertiary contact with the stem loop SL4 (**Figure 3A**). Lacking the extra stem loops in twister sister ribozyme, the key contacts for U38 of LINE-1-core (C8 in OR4K15-core) are the H-bond interaction with Al 1 N6 (A41 N6 in OR4K15-core), and the interactions between the phosphate and G7 (G37 in OR4K15-core). No involvement of the nucleobase part (U38 in LINE-l-core, C8 in OR4K15-core) in the modelled structures is compatible to the mismatches U38C and U38A found in the comparison,

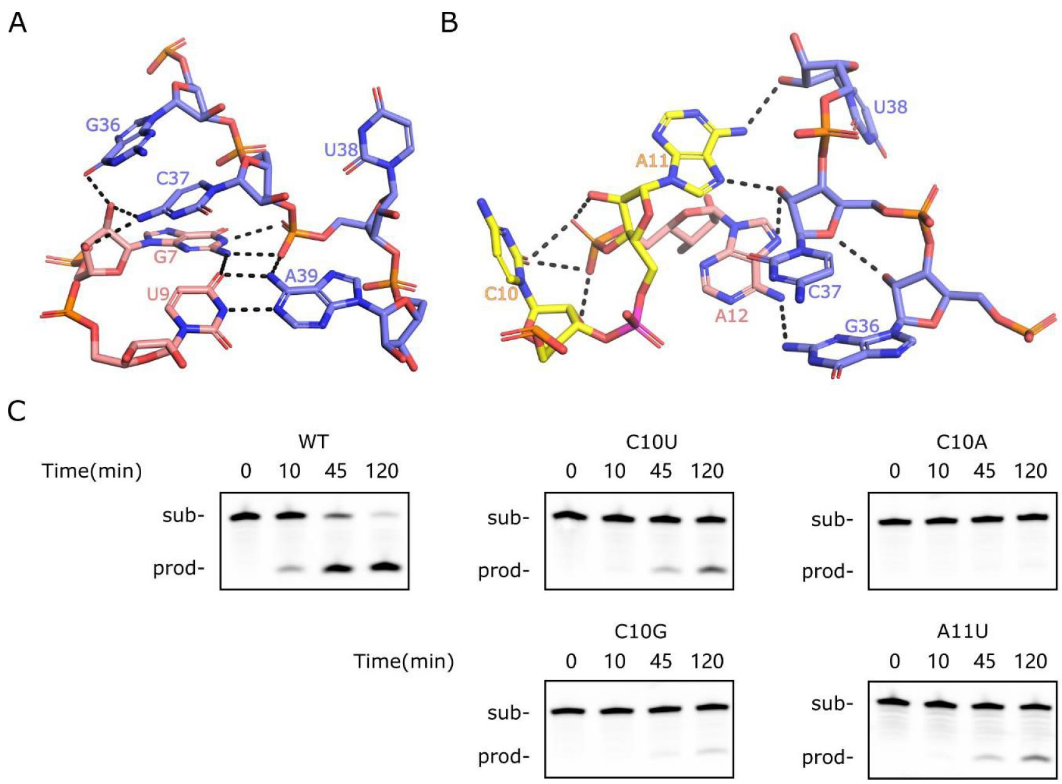


Figure 4.

Special features of the homology modelled LINE-1-core structure.

(A) The network of H-bonding interactions involving G7. (B) Intermolecular contacts at the Cl O-All cleavage site in the modelled structure. The scissile phosphate is colored magenta. (C) PAGE analysis of the cleavage activity of wild-type (WT) and mutants C10U, C10G, C10A, and A11U.

as changes of the nucleobase part did not have a big influence on the H-bonded interactions involving U38 (C8 in OR4K15-core) (**Figure 3A**). Thus, it is not entirely clear whether the model structure of TS-like ribozymes built from twister sister will remain stable in the absence of any tertiary interactions with the stem loop SL2 or it will deviate significantly from the twister sister.

Differences in mutational responses in the Lis between the TS-like and the twister sister ribozymes.

Considering the high similarity of the internal loops, we further investigated the mutational effects on the internal loop Lis. Most nucleotides in the LI region of LINE-l-rbz are relatively conserved, as single mutations showed RA' < 0.2 according to deep mutational scanning results (**Figure 2A**). The conservation is consistent with the stacking and hydrogenbonding interactions in the catalytic region. By comparison, mutations of C62 (C54 in three-way junctional twister sister ribozyme) at the cleavage site did not lead a major change on the cleavage activity in previous studies (**32**, **33**). This can be compared to the fact that single mutations of the corresponding nucleotide in LINE-l-rbz (CIO) had relative activities as low as 0.07 in the deep mutational scanning result. To further confirm the above result, we performed cleavage assays on several mutations at the cleavage site of LINE-l-core. As shown in **Figure 4C**, mutations on CIO showed either partial (C10U, C10G) or complete loss (C10A) in cleavage activity. While the corresponding mutations in the four-way junctional twister sister ribozyme showed pronounced (C62U, C62A) or somewhat reduced (C62G) cleavage activity (**32**, **33**). More interestingly, A11U in LINE-l-core showed partial cleavage activity, whereas the corresponding mutation A63U in the four-way junctional twister sister ribozyme showed complete loss of cleavage activity (**32**, **33**). These different mutation effects indicate that LINE-l-core may not adopt exactly the same structure and catalytic mechanism as twister sister ribozyme. In other words, TS-like ribozymes may not be simply a minimal version of twister sister ribozymes.

We further examined the interactions around the cleavage site C 10-Al 1 in LINE-l-rbz (C40- A41 in OR4K15-core). The corresponding nucleotides in twister sister ribozymes (C62-A63 in four-way junctional twister sister ribozyme, C54-A55 in three-way junctional twister sister ribozyme) adopt either a splayed-apart or a base-stacked conformation, with the scissile phosphate H-bonded to the G5 (four-way junctional twister sister ribozyme) or C7 (three-way junctional twister sister ribozyme). In our modelled structures (**Figure 4B**, **Supplementary Figure S9D**), the bases at the cleavage site CIO-All in LINE-l-core (C40-A41 in OR4K15- core) are splayed away, with All (A41 in OR4K15-core) directed inwards and CIO (C40 in OR4K15-core) directed outwards. In addition, H-bonded interactions between CIO (C40 in OR4K15-core) and the two nucleotides 3' of CIO could be found (**Figure 4B**, **Supplementary Figure S9D**). These structure differences may partially explain why mutations of CIO were detrimental to the catalysis in TS-like ribozymes. Moreover, we did not observe interactions between CIO (C40 in OR4K15-core) and G36/C37 (G6/C7 in OR4K15-core) although they were important to anchor the cytosine in twister sister ribozymes. This may be due to replacement of the non-canonical pair G5U64 (G6U57 in three-way junctional twister sister ribozyme) by A12G36 (G6A42 in OR4K15-core) in TS-like ribozymes. Thus, the stability of the current model structure based on the template from twister sister is uncertain.

Activity confinnation and biochemical analysis of LINE-l-core and OR4K15-core.

To confirm the cleavage activity of the core region (without the stem-loop linker), we obtained the segments (54–71 for LINE-1, 101–116 for OR4K15) as the substrate strands with the cleavage site and the segments (83–99 for LINE-1, 70–84 for OR4K15) as the enzyme strands (**Figure 5A**), as they made up the LINE-l-core and OR4K15-core. When the substrate strand was mixed with the enzyme strand, most of the substrate strand was cleaved in the presence of Mg^{2+} (**Figure 5B**). When Mg^{2+} was not included in the reaction, no cleavage could be observed even after 24h's incubation (**Supplementary Figure S10**). This result confirms that the RNA cleavage in TS-like ribozymes is a catalytic reaction, accelerated by Mg^{2+}. Moreover, the result confirmed that the stem loop SL2 regions in LINE-l-rbz and OR4K15-rbz (**Figure 2**) did not participate in the catalytic activity.

The core constructs of the two TS-like ribozymes contain 35 and 31 nucleotides, respectively. They are the simplest and the second simplest self-cleaving ribozymes reported so far. In all previously reported self-cleaving ribozyme families, the self-cleaving reaction occurs through an internal phosphoester transfer mechanism, in which the 2'-hydroxyl group of the -1 (relative to the cleavage site) nucleotide attacks the adjacent phosphorus resulting in the release of the 5' oxygen of+1 (relative to the cleavage site) nucleotide (**2**, **34**–**37**). We confirmed that both LINE-l-core and OR4K15-core employed the same cleavage mechanism because the analogous RNAs that lacked the 2' oxygen atom in the -1 nucleotide (dClO for LINE-l-core, dC40 for OR4K15-core) were unable to cleave (**Figure 5B**). We also investigated whether TS-like ribozymes can cleave when Mg^{2+} was replaced by $Co(NH_3)_6{}^{3+}$ in the reaction. $Co(NH_3)_6{}^{3+}$ is isosteric with $Mg(H_2O)_6{}^{2+}$, but the divalent cations cannot directly participate in catalysis as the amino ligands cannot readily dissociate. In **Figure 5C**, a total loss of cleavage activity in $Co(NH_3)_6{}^{3+}$ in the absence of Mg^{2+} indicates that the TS-like ribozymes require the binding of divalent cations not only for structure folding, but also for catalysis.

As shown in **Figure 5D**, we further examined the dependence of the metal ions on TS-like ribozymes' cleavage activity. At a concentration of ImM, ribozyme cleavage can be observed with Mg^{2+}, Mn^{2+}, Co^{2+} and Zn^{2+} but little or none with Ca^{2+}, Cu^{2+}, Ba^{2+}, Ni^{2+}, Na^+, K^+, Li^+, Cs^+ or Rb^+, indicating that direct participation of specific hydrated divalent metal ions is required for self-cleavage. More interestingly, we found that the TS-like ribozymes have an equivalent or even higher cleavage ratio with Mn^{2+} than Mg^{2+}. For LINE-l-core, the cleaved fractions were ~57% for Mg^{2+}, ~74% for Mn^{2+}. For OR4K15-core, the cleaved fractions were ~9% for Mg^{2+}, ~79% for Mn^{2+}. We further characterized the cleavage rates of these two ribozymes under different concentrations of Mg^{2+} and Mn^{2+} by using a fluorescence resonance energy transfer (FRET)-based method, as shown in **Supplementary Figure S11** and **Figure S12**. We used the first-order rate constant (k_{obs}) to represent the efficiency of the cleavage reaction. The k_{obs} of LINE-l-core under single-turnover condition was ~0.05 min^1 when measured in lOrnM $MgCl_2$ and 100mM KCl at pH 7.5 (**Figure S13**). Only a slightly lower value of k_{obs} (~0.03 min^{-1}) was observed for LINE-l-rbz (**Figure S14**). This confirms that the stem loop region SL2 does not contribute much to the cleavage activity of the TS-like ribozymes. Cleavage activities of the two TS-like ribozymes were highly dependent on the concentration of divalent metal ions. The steep increase in rate constants of two TS-like ribozymes plateaued at Mg^{2+} concentrations above 100mM, while for Mn^{2+} it plateaued at concentrations of 10mM. Consistent with the PAGE result, LINE-l-core showed higher cleavage rates than OR4K15- core when they were incubated with Mg^{2+}. However, the cleavage rates of OR4K15-core were only lower than LINE-l-core when the concentrations of Mn^{2+} were less than 20mM, but higher, otherwise. Thus, there may exist important difference between OR4K15-core and LINE-l-core to explain this different bias toward ions. A detailed explanation for this difference may require high-resolution structure determination of the TS-like ribozymes.

Homology search of two TS-like ribozymes.

To locate close homologs of the two TS-like ribozymes, we performed cmsearch based on a covariance model (**38**) built on the sequence and secondary structural profiles. In the human genome, we got 1154 and 4 homolog sequences for LINE-l-rbz and OR4K15-rbz, respectively. For OR4K15-rbz, there was an exact match located at the reverse strand of the exon of OR4K15 gene (**Figure 6A**). The other 3 homologs of OR4K15-rbz belongs to the same olfactory receptor family 4 subfamily K (**Figure 6C**). However, there was no exact match for LINE-l-rbz (**Figure 6A**). Interestingly, a total of 1154 LINE-l-rbz homologs were mapped to the LINE-1 retrotransposon according to the RepeatMasker (*http://www.repeatmasker.org*) annotation. **Figure 6B** showed the distribution of LINE-l-rbz homologs in different LINE-1 subfamilies in the human genome. Only three subfamilies L1PA7, L1PA8 and L1P3 (L1PA7-9) can be considered as abundant with

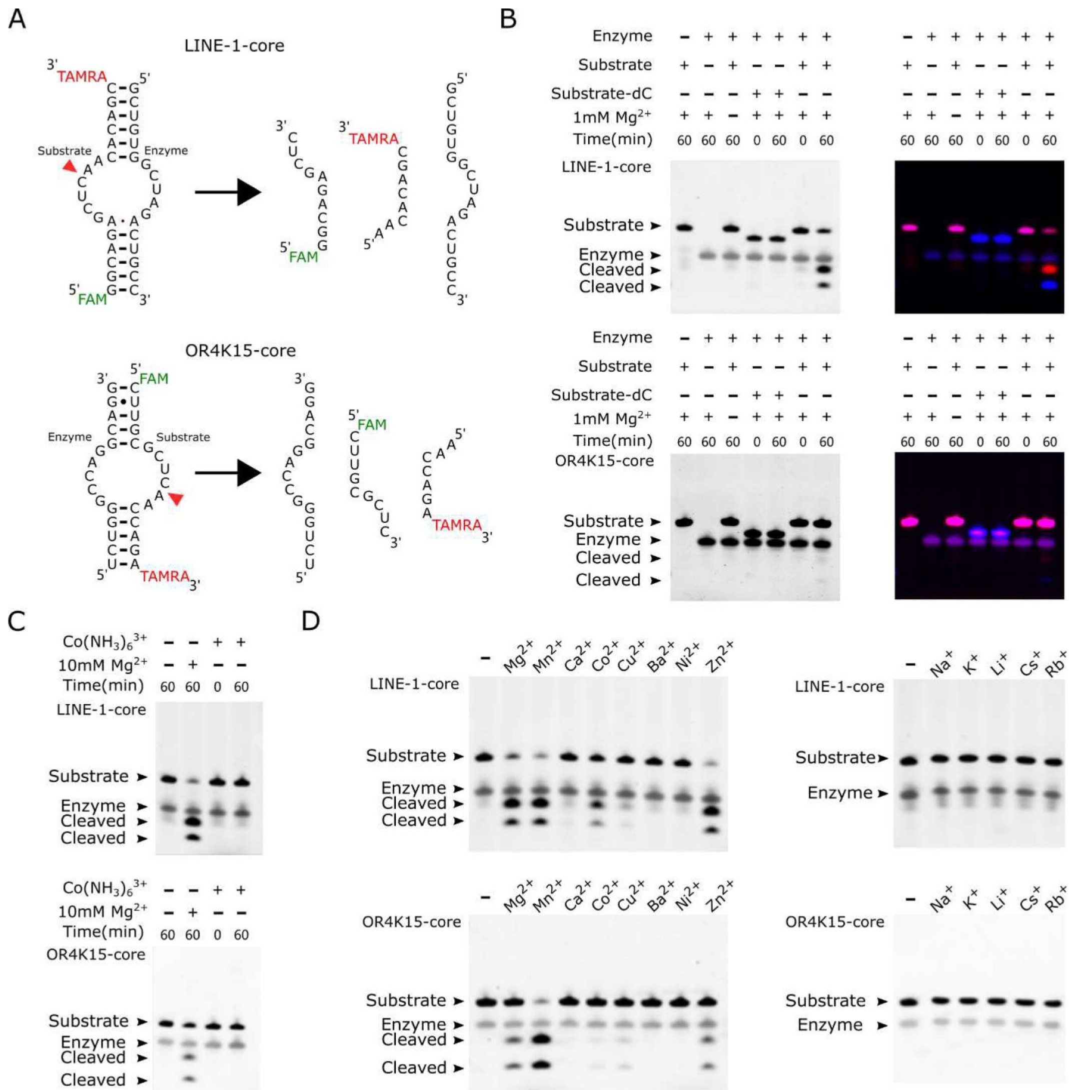


Figure 5.

(A) (Top) The components of the bimolecular construct of LINE-l-core during cleavage, substrate strand (54–71) and enzyme strand (83–99). (Bottom) The components of the bimolecular construct of OR4K15-core during cleavage, the enzyme strand (70–84) and substrate strand (101–116). The red arrowhead indicates the cleavage site. (B) PAGE-based cleavage assays of the bimolecular construct with the all-RNA substrate and a substrate analog (dC) wherein a 2’-deoxycytosine is substituted for the cytosine ribonucleotide at position 10 or 40 of the substrate RNA. The substrate RNA was incubated with the enzyme RNA at 37°C for lh in the presence (+) or absence (-) of ImM $MgCl_2$. The single-channel fluorescent images (left) were generated by using UV excitation (302 nm) and 590/110 nm emission on the ChemiDoc MP imaging system. The multi-channel fluorescent images (right) are overlays of two scans. They were generated from the ChemiDoc MP imaging system (Bio-Rad), Fluorescein (excitation: Epi-Blue 460–490 nm, emission: 532/28) for FAM, Cy3 (excitation: Epi-Green 520–545 nm, emission: 602/50) for TAMRA. (C) Cleavage assays in the absence (-) or presence of cobalt hexammine chloride [$Co(NH_3)6Cl_3$] or $MgCl_2$ at 5mM for lh. (D) Cleavage assays at 37°C for Ih, in the absence (-) or presence (+) of various metal ions. Divalent metal ions (left) were used at a final concentration of ImM, and monovalent metal ions (right) were used at a final concentration of IM.

LINE-l-rbz homologs (>100 homologs per family). The consensus sequences of all homologs obtained are shown in **Figure 6D**. In order to investigate the self-cleavage activity of these homologs, we mainly focused on the mismatches in the more conserved internal loops. The major differences between the 5 consensus sequences are the mismatches in the first internal loop. The widespread A12C substitution can be found in majority of LINE-l-rbz homologs, this substitution leads to a one-base pair extension of the second stem (P2) but almost no activity (RA': 0.03) based on our deep mutational scanning result. Then we selected 3 homologs without A12C substitution for LINE-l-rbz for in vitro cleavage assay (**Figure 6E**). But we didn't observe significant cleavage activity, this might be caused by GU substitutions in the stem region. For 3 homologs of OR4K15-rbz, we only found one homolog of OR4K15 with pronounced self-cleavage activity (**Figure 6F**). In addition, we performed similar bioinformatic search of the TS-like ribozymes in other primate genomes. Similarly, the majority (15 out of 18) of primate genomes have a large number of LINE-1 homologs (>500) and the remaining three have essentially none. However, there was no exact match. Only one homolog has a single mutation (U38C) in the genome assembly of Gibbon (**Figure S15**). The majority of these homologs have 3 or more mismatches (**Figure S15**). For OR4K15-rbz, all representative primate genomes contain at least one exact match of the OR4K15-rbz sequence.

Discussion

The wide distribution of RNA self-cleaving activity in different species suggest that selfcleaving ribozymes play an important role in living organisms. Most self-cleaving ribozymes (**39**) are in non-coding regions of the hosting genome. For example, the CPEB3 ribozyme in the intron of the CPEB3 gene (**8**), hammerhead ribozyme in the 3' UTR of Clec2 genes (**9**), hammerhead and HDV motifs associated with retrotransposons (**11**, **40**). LINE-1 belongs to the family of retrotransposons, they make up about 17% of the human genome, with over 500,000 copies (**41**, **42**). The retrotransposition process of LINE-1 increases the copy number of repeats and generates insertional mutations in the genome, which contributes to the genetic novelties as well as genomic instability. However, we found that the LINE-l-rbz sequence shared at least two mismatches with the ancient human LINE-1 families, that led to the loss of self-cleavage activity. Considering the location of the LINE-1 ribozyme inside the 5' UTR, it is suggested that this ribozyme might be related to the transcriptional regulation of the LINE-1 retrotransposon during evolution, like the HDV-like ribozymes. By comparison, the OR4K15 ribozyme is located at the reverse strand of the coding region of the olfactory receptor gene OR4K15, suggesting a potentially novel functional mechanism for this ribozyme as antisense transcription has been shown to be ubiquitous in mammals and may affect the gene regulation (**43**). However, the progress on understanding the functional roles of LINE-1 and OR4K15 self-cleaving ribozymes was slow because of lacking structural clues.

The centerpiece of structural clues is the RNA base-pairing structure resulted from the complex interplay of secondary and tertiary interactions. Previously, we developed a technique (**26**) for inferring the base-pairing information of the self-cleaving ribozymes by deep mutational scanning. In this paper, we applied this technique to the LINE-1 and OR4K15 ribozymes, to obtain the functional regions and the base-pairing information. As both LINE-1 and OR4K15 ribozymes were discovered from the randomly fragmented human genomic DNA selection based biochemical experiments, the original full-length (WT) sequences of the ribozymes do not necessarily represent the functional region of the RNAs. However, the functional regions can cleave when they were transcribed along with the upstream and downstream sequences, and, thus, they are more likely to be active in cellular environment.

Indeed, we found that the functional regions for both ribozymes are very short (35 and 31 nucleotides for LINE-1 and OR4K15, respectively) and surprisingly in the same class in a circular permutated form. We are confident about the secondary structures obtained because of the

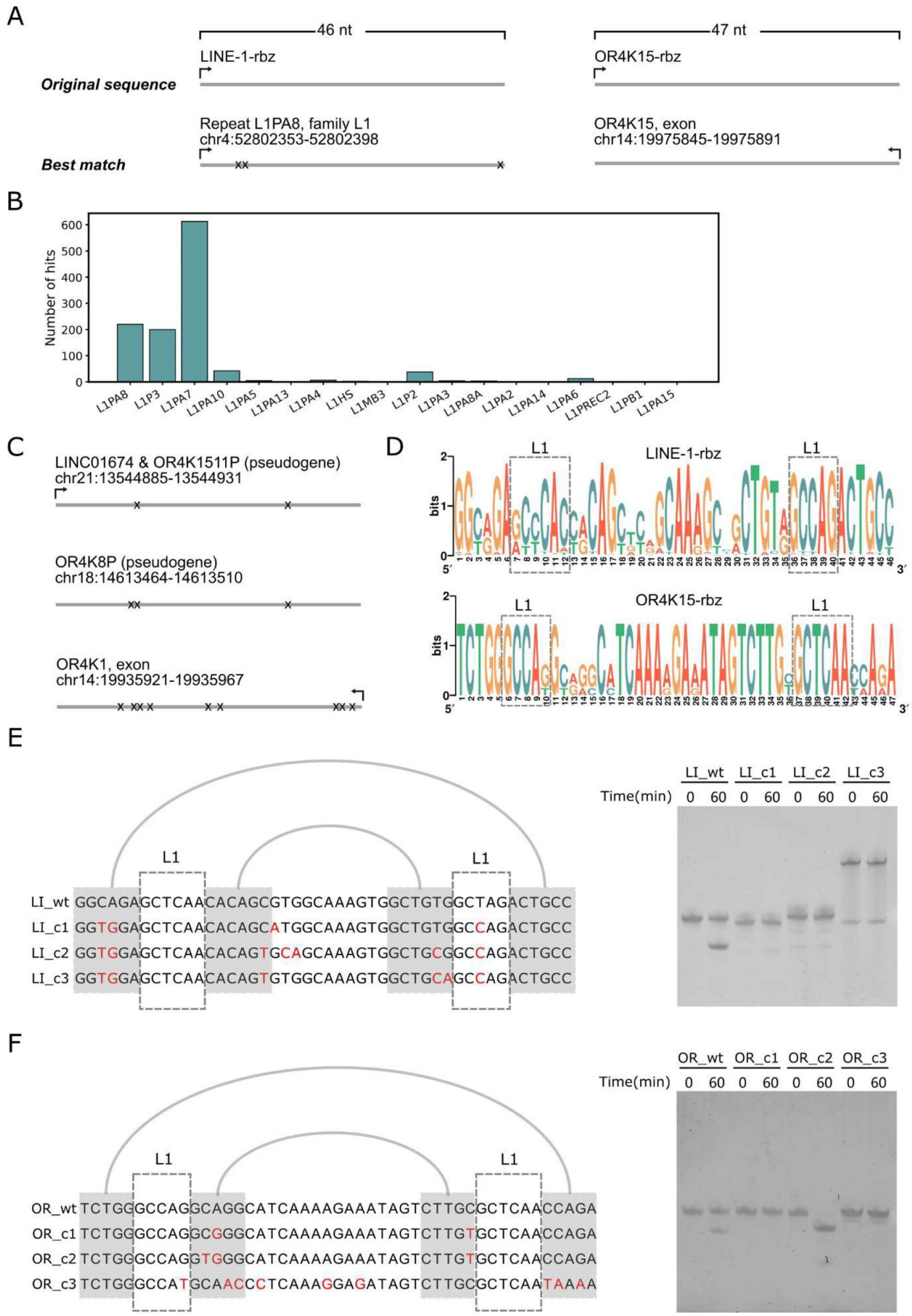


Figure 6.

(A) Genomic locations of the sequences with the highest similarity to the LINE-1- rbz (left) and OR4K15-rbz (right). (B) The distribution of LINE-1-rbz homologs in different LINE-1 subfamilies of the human genome assembly (hg38). (C) Genomic locations of OR4K15-rbz homologs. (D) Nucleotide compositions of LINE-l-rbz homologs found in the human genome. (E) Sequence alignment (left) and PAGE result (right) of 4 LINE-l-rbz homologs. (F) Sequence alignment (left) and PAGE result (right) of 4 OR4K15-rbz homologs. OR_cl, OR_c2, OR_c3 correspond to the 3 homologs (from top to bottom) shown in (C).

consistency between the results from the deep mutational study of the original sequence and the functional region of the LINE-1 ribozyme. The obtained secondary structure is further confirmed by the discovery of the functional regions of the two TS-like ribozymes in the circular permutated form from one another (**Figure 3A**). From the mutational information of LINE-l-rbz, the only mismatch (U38C) from OR4K15 in the catalytic region has the RA' of 0.6. While the corresponding mismatch mutation C8U in OR4K15 has the RA of 1.54, or 0.65 for U8C. This consistence in activity reduction for the U-→C mutation further confirmed that these two ribozymes belong to the same class in a circular permutated form. Currently, three classes of self-cleaving ribozymes, the hammerhead (**7**, **44**—**47**), twister (**34**) and the hairpin (**48**) ribozymes are found in different species in multiple circular permutated forms. The discovery of permutated TS-like ribozymes in the human genome for the first time suggests that we can find additional examples of circular permutated TS-like ribozymes in other species.

More interestingly, we found that the internal loop regions of the TS-like ribozymes share a high similarity with the known catalytic region of the twister sister ribozyme (2 mismatches only, **Figure 3A**). The conserved cleavage sites suggest a similar structure and mechanism as shown in **Figure 3A**. However, the maximum observed rate constant (k_{obs}) for twister sister ribozyme was ~5 min^{-1} (**37**), compared to ~0.ll ± 0.01min^{1} for the TS-like ribozymes. This rate difference is not mainly caused by mismatches in the catalytic core between the twister sister and LINE-1/OR4K15 ribozymes. In fact, deep mutational scanning results indicate that one mismatch (U38C) in the catalytic region of OR4K15-rbz (**Figure 3A**) has the RA' of 0.6. Two mismatches (A12U and U38A) in the catalytic region of the twister sister ribozyme (**Figure 3A**) have the RA' of 0.28 and 0.47, respectively, whereas the coexistence of these two mismatches only has the RA' of 0.09. Thus, a more sophisticated structure along with long-range interactions involving the SL4 region in the twister sister ribozyme must have helped to stabilize the catalytic region for the improved catalytic activity. Similarly, previous studies have demonstrated that peripheral regions of hammerhead (**49**), hairpin (**50**) and HDV (**51**,**52**) ribozymes could greatly increase their self-cleavage activity. Given the importance of the peripheral regions, absence of this tertiary interaction in the TS-like ribozyme may not be able to fully stabilize the structural form generated from homology modelling.

Indeed, we found some differences between the responses to mutations in the catalytic regions around the cleavage sites of TS-like ribozymes and those of twister sister ribozymes. Although there are pronounced differences around the cleavage sites in previous twister sister ribozyme structures, both cytosines at the cleavage sites were found to be insensitive to mutations (**32**, **33**). However, unlike twister sister ribozymes, the cytosine at the cleavage site of TS-like ribozyme was sensitive to mutations in our mutation assays. According to our modelled structures, the H-bonding interactions associated with the cleavage site C10-A11 (C40-A41 in OR4K15-core) in TS-like ribozymes show that the cytosine (CIO in LINE-l-core, C40 in OR4K15) is relatively conserved, which is consistent with the mutation assay results. In the modelled structure, we did not observe direct contacts helped to stabilize the scissile phosphate and the in-line alignments required for the catalytic step. This may suggest that conformational changes may be required to generate the active conformation prior to the catalytic step as in the twister sister ribozyme structures. Or this is due to inexact structures generated from homology modelling, because Mg^{2+} was not included during homology modelling. Our experimental results (**Figure 5B-D**) showed that Mg^{2+} is essential for both structure folding and catalysis. Absence of Mg^{2+} during modelling might lead to the conformational change of the tertiary structures. Thus, further studies are clearly needed to illustrate possible structural and mechanistic difference between TS-like ribozymes and twister sister ribozyme.

According to the bioinformatic analysis result, there are some TS-like ribozymes (one LINE-1-rbz homolog in the Gibbon genome, and some OR4K15-rbz homologs) with in vitro cleavage activity in primate genomes. Unlike the more conserved CPEB3 ribozyme which has a clear function, the function of the TS-like ribozymes is not clear, as they are not conserved, belong to the pseudogene or located at the reverse strand. While the *in vivo* activities and functions of these two circular-permutated TS-like ribozymes require further studies, their discovery may have important therapeutics and biotech implications. First, the TS-like ribozymes have the simplest secondary structure with two stems and two internal loops, compared to all naturally occurring self-cleaving ribozymes. Previous studies also have revealed some minimized forms of self-cleaving ribozymes, including hammerhead (**19**, **53**) and HDV-like (**54**) ribozymes. However, when comparing the conserved segments, they (>= 36 nt) are not as short as the TS-like ribozymes (31 nt) found here. Second, this TS-like ribozyme after removing the stem-loop region only requires a 15-nucleotide enzyme strand and a 16-nucleotide substrate strand for its function. Thus, it can be easily modified into a trans-cleaving ribozyme, which was found useful for bioengineering and RNA-targeting therapeutics (**55**). A recent intracellular selectionbased study has shown that hammerhead ribozyme can be engineered into a trans-acting ribozyme with consistent gene knockdown ability on different targeted mRNA either in prokaryotic or eukaryotic cells (**55**). The TS-like ribozyme, the simplest one known, may have an unprecedented advantage, compared to existing trans-cleaving ribozymes.

What is also important is that this work further confirms the usefulness of deep mutational scanning and high-throughput sequencing for RNA structural inference by CODA analysis (**26**, **56**). The method is not limited to self-cleaving ribozymes, if combined with other functional or phenotypic selection techniques. Moreover, the structural and full mutational information provided by this method could be utilized to discover additional functional RNAs and thus help us to investigate the hidden side of non-coding RNAs.

Materials and Methods

All the oligonucleotides listed in **Supplementary Table S4**, except for the doped library, were purchased from IDT (Integrated DNA technologies) and Genscript. The doped mutant library Rz_LINE-l_doped with a doping rate of 6%, was purchased from the Keck Oligo Synthesis Resource at Yale University. All the high-throughput sequencing experiments were performed on the Illumina HiSeq X platform by Novogene Technology Co., Ltd.

Deep mutational scanning experiments on original ribozyme sequences

The workflow for two deep mutational scanning experiments is illustrated in **Supplementary Figure S13**. The deep mutational scanning of the original sequences of LINE-1 ribozyme (labelled as LINE-l-ori) and OR4K15 ribozyme (labelled as OR4K15-ori) followed the same protocol as our previous work (**26**). Briefly, the mutant library was generated from 3-rounds of error-prone PCR, barcoded using primer Bar F and Bar R (**Supplementary Table S4**), and then diluted. After that, the transcribed RNA of the barcoded mutant library was reverse transcribed with RT ml3f adpl and template switching oligo TSO (**Supplementary Table S4**). The total cDNA was used to generate the RNA-seq library with primer P5R1 adpl and P7R2 adp2 (**Supplementary Table S4**). The DNA-seq libraries were generated by amplifying the ribozyme mutant library with primer P5Rl_ml3f and P7R2_t7p (**Supplementary Table S4**). Both DNA-seq and RNA-seq libraries were sequenced on an Illumina HiSeq X sequencer with 25% PhiX control by Novogene Technology Co., Ltd. We counted the number of reads of the cleaved and uncleaved portions of each variant by mapping their respective barcodes. The relative activity (RA) of each variant is calculated by using the equation $RA(var) = N_{cleaved}(var)\ N_{total}(wt)/N_{total}(var)\ N_{cleaved}(wt)$.

Covariation-induced deviation of activity (CODA) analysis

We used CODA (**26**) to analyze the RA of LINE-l-ori and OR4K15-ori. For Monte Carlo (MC) simulated annealing, we used the same weighting factor of 2 for both datasets as employed previously (**26**).

Deep mutational scanning and CODA analysis on the functional region of LINE-1 ribozyme

We identified the functional region of the LINE-1 ribozyme (labelled as LINE-l-rbz) by removing the terminal sequence positions whose mutations do not affect cleavage activities. Then, the second round of deep mutational scanning experiment was applied to the LINE-l-rbz. Here, we used a doped mutation library instead of a library generated by error-prone PCR. The doped mutant library Rz_LINE-l_doped (**Supplementary Table S4**) was amplified by using primer T7prom and M13F (**Supplementary Table S4**). The PCR product was then gel-purified, quantified, and diluted. Approximately 5×10^5 doped DNA molecules were amplified by T7prom and M13F primers (**Supplementary Table S4**) to produce enough DNA template for in vitro transcription. The transcribed and purified RNAs (10 pmol) of the mutant library were mixed with 100 pmol rM13R_5desBio_3P (**Supplementary Table S4**), 2 μl 10X RtcB reaction buffer (New England Biolabs), 2 μl 1mM GTP, 2 μl 10mM $MnCl_2$, 0.2 μl of Murine RNase Inhibitor (40 U/μl, New England Biolabs) and 1 μl RtcB RNA ligase (New England Biolabs) to a total volume of 20 μl. The reaction mixture was incubated at 37 °C for 1 h, then purified using RNA Clean & Concentrator-5 kit (Zymo Research). The purified products were mixed with 2 μl of 10 μl M RT_ml3f_ adp1 (**Supplementary Table S4**) and 1 μl of 10mM dNTPs in a volume of 8 μl, and then heated to 65 °C for 5min and placed on ice. Reverse transcription was initiated by adding 4 μl of 5x ProtoScript II Buffer (New England Biolabs), 2 μl of 0.1 M DTT, 0.2 μl of Murine RNase Inhibitor (40 U/pl, New England Biolabs) and 1 μl ProtoScript II RT (200 U/μl, New England Biolabs) to a total volume of 20 μl. The reaction mixture was incubated at 42°C for 1 h, and then purified by Sera-Mag Magnetic Streptavidin-coated particles (Thermo Scientific).

The purified products after streptavidin-based selection were amplified by PCR to construct the RNA-seq library with primer P7R2_ml3r and P5Rl_ml3f (**Supplementary Table S4**). The DNA template used for transcription was used as the template for constructing the DNA-seq library with primer P5Rl_ml3f and P7R2_t7p (**Supplementary Table S4**). The DNA-seq and RNA-seq libraries were also sequenced on an Illumina HiSeq X sequencer with 25% PhiX control by Novogene Technology Co., Ltd. Raw paired-end sequencing reads were filtered and then merged by using the bioinformatic tool PEAR (**57**) to generate the high-quality merged reads. We counted the read number of each variant in DNA-seq and RNA-seq by mapping the barcode. Then, we estimated the relative activity (RA') of each variant by using the equation $RA'(var) = N_{RNAseq}(var)N_{DNAseq}(wt)/N_{DNAseq}(var)N_{RNAseq}(wt)$.

This relative activity (RA') measure is different from the relative activity (RA) when both cleaved and uncleaved sequences were obtained in RNA-seq in the first-round deep mutational scanning. However, it should be a good estimate of RA because a mutant with a higher activity (cleavage rate) should have a higher possibility to be captured by RtcB ligase. We confirmed RA' as a good estimate of RA by comparing shared single mutations from LINE-l-ori and LINE-l-rbz and obtained a strong Pearson correlation coefficient of 0.66 (Spearman correlation coefficient of 0.769). Moreover, both RA and RA' (**18**–**30**) show that the central region of LINE-l-rbz is insensitive to mutations (**Supplementary Figure S5**). Moreover, secondary structures derived from RA' and RA are consistent with each other (see below), confirming the correct identification of structural and functional regions of LINE-1 ribozyme.

We used the same method CODA (**26**) to analyze the RA' of LINE-l-rbz. For Monte Carlo (MC) simulated annealing, we used the same weighting factor of 2.

PAGE-based cleavage assays

Both ribozyme sequences were separated into two portions: the substrate part (LI_S_5F3T and OR_S_5F3T, **Supplementary Table S4**) and the enzymatic part (LI_E and OR_E, **Supplementary Table S4**). LI_S_5F3T and OR_S_5F3T were synthesized and HPLC purified by Genscript. Cytosine CIO and C40 were substituted by deoxy cytosine in LI_S_dC10 and OR_S_dC40 (**Supplementary Table S4**) during synthesis.

In a 20 µl reaction system, 40 pmol LI_S_5F3T or OR_S_5F3T was mixed with 2 µl 0.2 M Tris-HCl and 2 µl 10 mM metal ion stock solution unless noted otherwise. 60 pmol LI_E/OR_E (**Supplementary Table S4**) was added to initiate the reaction. Reactions were incubated at 37°C for 1 h, and then stopped by adding an equal volume of stop solution (95% formamide, 0.02% SDS, 0.02% bromophenol blue, 0.01% Xylene Cyanol, ImM EDTA). The reaction products were then separated by denaturing (8 M urea) 15% PAGE and visualized by SYBR Gold (Thermo Fisher) staining.

For bimolecular mutants of LINE-l-rbz and OR4K15-rbz, we separated the ribozymes into the substrate part (5’ 6-FAM labelled) and the enzymatic part in a similar way. In a 20 µl reaction system, 40 pmol substrate RNA was mixed with 2 µl 0.2 M Tris-HCl and 2 µl 10 mM Mg^{2+} stock solution. 60 pmol enzyme RNA was added to initiate the reaction. Reactions were incubated at 37°C for different time intervals, and then stopped by adding an equal volume of stop solution (95% formamide, 0.02% SDS, 0.02% bromophenol blue, 0.01% Xylene Cyanol, ImM EDTA). The reaction products were then separated by denaturing (8 M urea) 15% PAGE.

For single-stranded mutants of LINE-l-rbz and OR4K15-rbz, we incubated 60 pmol RNA in a 20 µl reaction system. Mg^{2+} stock solution was added to initiate the reaction. The final concentrations for the reaction solution were 20 mM Tris-HCl buffer (pH 7.5), 100 mM KC1, and 10 mM MgCh. Reactions were incubated at 37°C for different time intervals, and then stopped by adding an equal volume of stop solution (90% formamide, 0.02% SDS, 0.02% bromophenol blue, 0.01% Xylene Cyanol, 50mM EDTA). The reaction products were then separated by denaturing (8 M urea) 12% PAGE.

Fluorescence-based kinetic analysis

The fluorescence-based kinetic analysis of the two TS-like ribozymes was similar to our previous work on the CPEB3 ribozyme. The substrate RNAs LI_S_5F3T and OR_S_5F3T were synthesized with 5’ FAM as the fluorophore and 3’ TAMRA as the quencher. Cleavage of the substrate part by the enzymatic part will relieve the quenching and therefore generate a fluorescence signal.

For one 100 µl reaction system, 100 pmol enzyme RNA was mixed with 20 pmol substrate RNA unless noted otherwise. In the ion-dependent assay, a final concentration of 20mM Tris-HCl solution (pH 7.5) and lOOmM KC1 was used. Different concentrations of MgCh or MnCh solutions were employed to initiate the reactions.

The fluorescence (Ex 488 nm/Em 520 nm) was measured at 37°C for 3 hours using BioTek Synergy Hl, and then normalized by the control (no divalent ions) at t = 0. The first-order rate constant of ribozyme cleavage k_{obs} was calculated in a similar way as previously described (**26**). The kinetics data were fitted to $F = A - Be^{-k_{obs}t}$, where F is the fluorescence intensity, t is the time, A is the fluorescence at completion, and B is the amplitude of the observable phase. Each data point in **Supplementary Figure S11** is the average of k_{obs} of 3 independent reactions.

3D structure modelling of the TS-like ribozymes

We matched the base-pairs of the TS-like ribozymes to the most similar twister sister ribozyme (PDB: 5Y87) by map_align (**58**). And then we modelled the RNA structure by replacing, inserting, and deleting the nucleotides with household RNA modelling python script. Finally, we use RNA-BRiQ (**59**) developed by our group to optimize the structure model. The source code can be accessed at *https://gitee.com/hongxu66/coda-map*.

Secondary structure-based search of LINE-l-rbz and OR4K15-rbz

All representative primate genome assemblies in fasta format and the corresponding RepeatMasker (*http://www.repeatmasker.org*) annotation files were downloaded from the UCSC genomic website (*http://genome.ucsc.edu*). We used cmbuild from Infernal (*60*, *61*) to build the covariance model (*38*) of LINE-l-rbz and OR4K15-rbz with the wild type sequence and predicted secondary structure from deep mutational scanning experiment as input. Afterwards, we used the covariance model to search against the latest release of different representative primate genome assemblies by using cmsearch from Infernal. The hits with E-value <1.0 were then used for downstream analysis.

Data Availability

Illumina sequencing data for the LINE-l-ori, OR4K15-ori and LINE-l-rbz were submitted to the NCBI Sequence Read Archive (SRA) under SRA accession number PRJNA662002 (*https://www.ncbi.nlm.nih.gov/sra/PRJNA662002*).

The homology modelling structures of LINE-l-core and OR4K15-core are available in ModelArchive (*https://modelarchive.org/doi/10.5452/ma-chp4j* for LINE-l-core; *https://modelarchive.org/doi/10.5452/ma-bqqf8_for_OR4K15-core*).

Competing financial interests

The authors declare no competing financial interests.

Additional Declarations:

The authors declare no competing interests.

Acknowledgements

This work was funded by China Postdoctoral Science Foundation (fund no. 2021M702295 to Z.Z.) and a grant from National Natural Science Foundation of China (Grant #22350710182) to Y.Z. This work was also supported in part by Australian Research Council DP180102060 and DP210101875 and by National Health and Medical Research Council (1121629) of Australia to Y.Z. We would like to acknowledge the use of the High-Performance Computing Cluster 'Gowonda' at Griffith University and the supercomputing facility at the Shenzhen Bay Laboratory to complete this research. This research/project has also been undertaken with the aid of the research cloud resources provided by the Queensland Cyber Infrastructure Foundation (QCIF) and Australian Research Data Commons (ARDC). We also gratefully acknowledge the support of NVIDIA Corporation with the donation of the Titan V GPU used for this research.

Author Contributions

JZ conceived of the study. JZ and JW designed experimental studies and wrote the manuscript; ZZ and JZ conducted experimental studies and wrote the manuscript; PX and XH developed computational methods. ZZ performed computational and bioinformatics analysis. YZ participated in the initial design, assisted in data analysis, and drafted the manuscript; and all authors read, contributed to the discussion, and approved the final manuscript.

Supplementary materials

Table S1.

The number of mutants and the coverage of single and double mutations of LINE-1 ribozyme in two deep mutational scanning experiments for the original sequence (LINE-1- ori) and the functional region (LINE-1-rbz), respectively.

Mutations	Single mutation			Double mutation		
	Number	Total	Coverage	Number	Total	Coverage
LINE-1-ori	429	438	97.9%	17671	95265	18.5%
OR4K15-ori	410	420	97.6%	14159	87570	16.2%
LINE-1-rbz	137	138	99.3%	9310	9315	99.9%

Table S2.

Summary of two deep sequencing results for the original sequence (LINE-1-ori) and the functional region (LINE-1-rbz), respectively.

	Number of DNA-seq reads	Number of Barcodes	Number of mutants	Number of RNA-seq reads	Cleaved reads	Uncleaved reads
LINE-1-ori	89053476	168221	70962	59208161	1982615	39512450
OR4K15-ori	184729367	215650	61515	104916717	4847896	83394992
LINE-1-rbz	66952586	373301	162752	69904460	69904460	NA

Table S3.

Sequences with similar motifs found in a specialized Rfam database.

Accession	Family	Start	End	Length
URS0000D6A998_12908/1-71	Twister-sister (RF02681)	3	67	71
Matched sequence: |GCAGG|GCTAG|GCCC|AGTCCCGTGCAAGCCGGGACCGCCCCGAAAGGGGCG|CGGC|GCTCAA|CCTGC|				
URS0000D698FC_12908/1-80	Twister-sister (RF02681)	3	76	80
Matched sequence: |CTAGG|GCAAG|GTAC|AATCCCGTGCAAGCCGGGATCAGGGGTAGGTGCGATCTACCCCTG|CTAC|GCTCAA|CCTAC|				
URS0000D6995E_12908/1-79	Twister-sister (RF02681)	3	75	79
Matched sequence: |CTAGG|GCAAG|GTAC|AATCCCGTGCAAGCCGGGATCAGGGGTAGTGCGATCTACCCCTG|CTAC|GCTCAA|CCTAC|				

Name	Sequence (5' - 3')	Notes
Rz_LINE-1_wt	TAATACGACTCACTATAGGGATACGTAGGTGGTTTTCC CCTGATAGTGCTAAGGGGGCCTGGAAGTTTGGACTGG GCAGAGCTCAACACAGCGTGGCAAAGTGGCTGTGGCT AGACTGCCTCTCTGGATTCCTCGTCACTCAGCAGGGCA TCTCTGAGAGAAATGCCATTTGAGTTACTCAGGATTAC TGGCCGTCGTTTTAC	DNA template.
Rz_OR4K15_ wt	TAATACGACTCACTATAGGGAGTTAACGGTCATAGGC CATGGAAACTAGGAGCACCATTTCGCTGCCAGTGAAG AAATGAACAAAGAAAATCTGGGCCAGGCAGGCATCAA AAGAAATAGTCTTGCGCTCAACCAGAAAGTCTGTAAT CATTTTAAGGGTAATTTGAGTTACTCAGGATTACTGGC CGTCGTTTTAC	DNA template.

Table S4.

Oligonucleotides used in this study.

Rz_LINE-1_doped	TAATACGACTCACTATAGGGGCAGAGCTCAACACAGCGTGGCAAAGTGGCTGTGGCTAGACTGCCAANNNNNNNNNNNNNNNNNNNACTGGCCGTCGTTTTAC	Doped library.
M13F	GTAAAACGACGGCCAGT	PCR primer.
T7prom	TAATACGACTCACTATAGGG	PCR primer.
RT_m13f_adp1	CCCTACACGACGCTCTTCCGATCTGTAAAACGACGGCCAGT	Reverse transcription primer.
TSO	CTCGGCATTCCTGCTGAACCGCTCTTCCGATCTrGrGrG	Template switching oligo.
rM13R_5desBio_3P	/5'deSBioTEG/rCrArCrArGrGrArArArCrArGrCrUrArUrGrArCrC/3'Phos/	Linker for RtcB ligation.
Bar_F	TAATACGACTCACTATAGGGA	PCR primer for adding barcode.
Bar_R	GTAAAACGACGGCCAGTNNNNNNNNNNNNNNNAATCCTGAGTAACTCAAAT	PCR primer for adding barcode.
P5R1_m13f	AATGATACGGCGACCACCGAGATCTACACTCTTTCCCTACACGACGCTCTTCCGATCTGTAAAACGACGGCCAGT	PCR primer for DNA-seq library preparation.
P7R2_t7p	CAAGCAGAAGACGGCATACGAGATCGGTCTCGGCATTCCTGCTGAACCGCTCTTCCGATCTTAATACGACTCACTATAGG	PCR primer for DNA-seq library preparation.
P5R1_adp1	AATGATACGGCGACCACCGAGATCTACACTCTTTCCCTACACGACGCTCTTC	PCR primer for RNA-seq library preparation.
P7R2_adp2	CAAGCAGAAGACGGCATACGAGATCGGTCTCGGCATTCCTGCTGAAC	PCR primer for RNA-seq library preparation.
P7R2_m13r	CAAGCAGAAGACGGCATACGAGATCGGTCTCGGCATTCCTGCTGAACCGCTCTTCCGATCTCAGGAAACAGCTATGACC	PCR primer for RNA-seq library preparation.
LI_S_5F3T	/5'6-FAM/rGrGrCrArGrArGrCrUrCrArArCrArCrArGrC/3'6-TAMRA/	Oligo for activity assay.
LI_E	rGrCrUrGrUrGrGrCrUrArGrArCrUrGrCrC	Oligo for activity assay.
LI_S_dC10	/5'6-FAM/rGrGrCrArGrArGrCrUCrArArCrArCrArGrC	Oligo for activity assay.
OR_S_5F3T	/5'6-FAM/rCrUrUrGrCrGrCrUrCrArArCrCrArGrA/3'6-TAMRA/	Oligo for activity assay.
OR_E	rUrCrUrGrGrGrCrCrArGrGrCrArGrG	Oligo for activity assay.

Table S4. (continued)

OR_S_dC40	/5'6-FAM/rCrUrUrGrCrGrCrUCrArArCrCrArGrA	Oligo for activity assay.
LI_S_5F	/5'6-FAM/rGrGrCrArGrArGrCrUrCrArArCrArCrArGrC	Oligo for activity assay.
LI_S_10A_5F	/5'6-FAM/rGrGrCrArGrArGrCrUrArArArCrArCrArGrC	Oligo for activity assay.
LI_S_10G_5F	/5'6-FAM/rGrGrCrArGrArGrCrUrGrArArCrArCrArGrC	Oligo for activity assay.
LI_S_10U_5F	/5'6-FAM/rGrGrCrArGrArGrCrUrUrArArCrArCrArGrC	Oligo for activity assay.
LI_S_11U_5F	/5'6-FAM/rGrGrCrArGrArGrCrUrCrUrArCrArCrArGrC	Oligo for activity assay.
LI_wt	rGrGrCrArGrArGrCrUrCrArArCrArCrArGrCrGrUrGrGrCrAr ArArGrUrGrGrCrUrGrUrGrGrCrUrArGrArCrUrGrCrC	Mutants for activity assay.
LI_c1	rGrGrUrGrGrArGrCrUrCrArArCrArCrArGrCrArUrGrGrCrAr ArArGrUrGrGrCrUrGrUrGrGrCrCrArGrArCrUrGrCrC	Mutants for activity assay.
LI_c2	rGrGrUrGrGrArGrCrUrCrArArCrArCrArGrUrGrCrArGrCrAr ArArGrUrGrGrCrUrGrCrGrGrCrCrArGrArCrUrGrCrC	Mutants for activity assay.
LI_c3	rGrGrUrGrGrArGrCrUrCrArArCrArCrArGrUrGrUrGrGrCrAr ArArGrUrGrGrCrUrGrCrArGrCrCrArGrArCrUrGrCrC	Mutants for activity assay.
OR_wt	rUrCrUrGrGrGrCrCrArGrGrCrArGrGrCrArUrCrArArArArGr ArArArUrArGrUrCrUrUrGrCrGrCrUrCrArArCrCrArGrA	Mutants for activity assay.
OR_c1	rUrCrUrGrGrGrCrCrArGrGrCrGrGrGrCrArUrCrArArArArGr ArArArUrArGrUrCrUrUrGrUrGrCrUrCrArArCrCrArGrA	Mutants for activity assay.
OR_c2	rUrCrUrGrGrGrCrCrArGrGrUrGrGrGrCrArUrCrArArArArGr ArArArUrArGrUrCrUrUrGrUrGrCrUrCrArArCrCrArGrA	Mutants for activity assay.
OR_c3	rUrCrUrGrGrGrCrCrArUrGrCrArArCrCrCrUrCrArArArGrGr ArGrArUrArGrUrCrUrUrGrCrGrCrUrCrArArUrArArArA	Mutants for activity assay.

Table S4. (continued)

Figure S1.

Mutation rates of two deep mutational scanning results at each nucleotide position of LINE-1 ribozyme.

The mutation rate was calculated as the proportion of mutations observed at each position for the DNA-seq library.

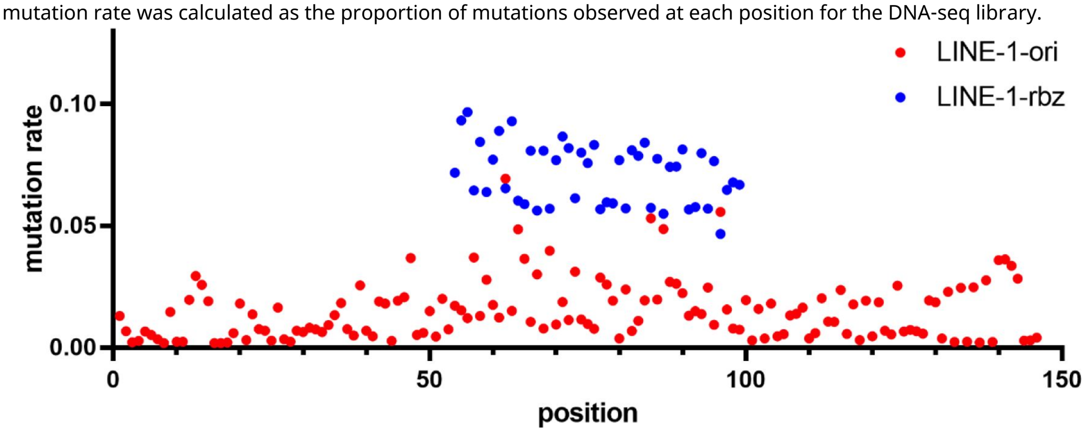


Figure S2.

Mutation rates at each nucleotide position of OR4K15 ribozyme.

The mutation rate was calculated as the proportion of mutations observed at each position for the DNA-seq library.

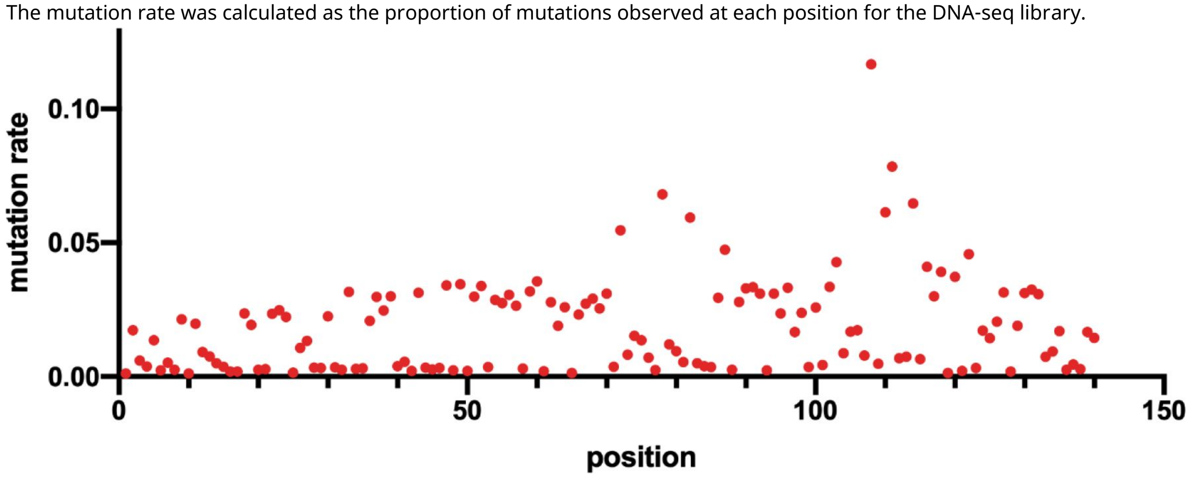


Figure S3.

Distribution of different mutation types in deep sequencing result of LINE-1 ribozyme.

The Y axis represents the ratio of each mutation type observed in all variants.

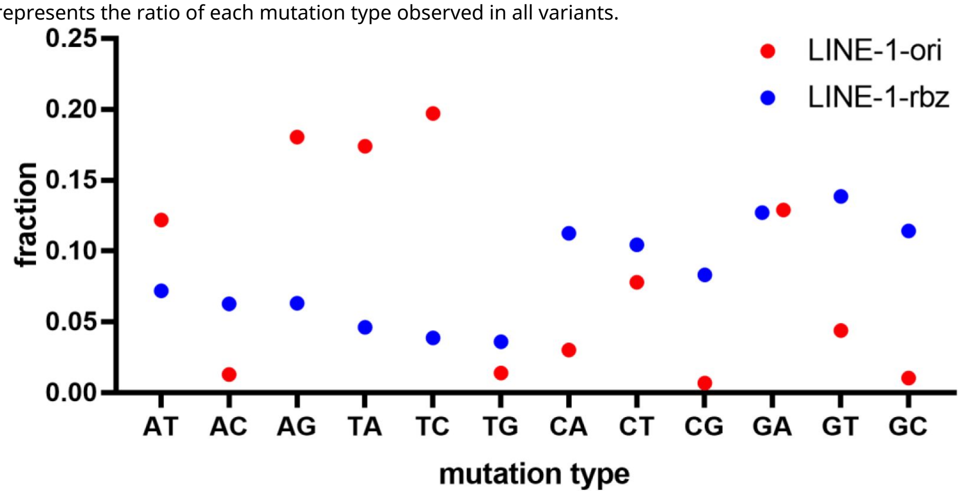


Figure S4.

Distribution of different mutation types in the deep sequencing result of OR4K15 ribozyme.

The Y axis represents the ratio of each mutation type observed in all variants.

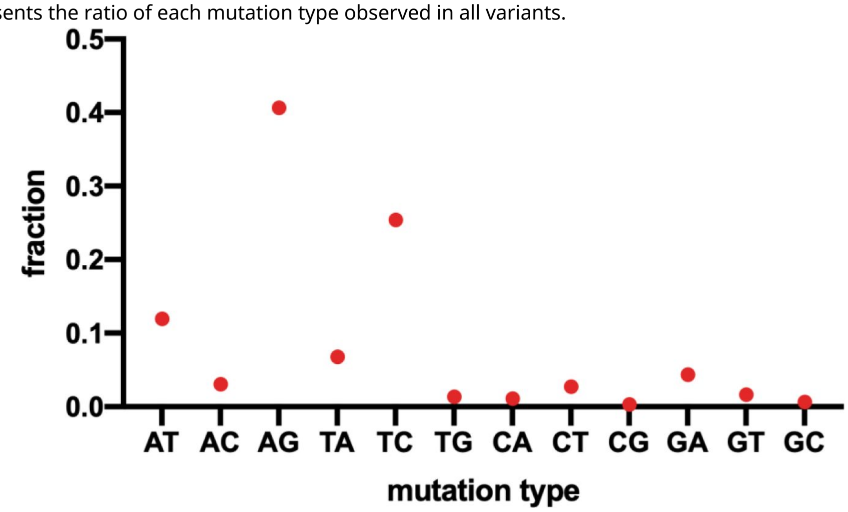


Figure S5.

Single mutation-based average RA’ and RA distribution. Blue, average RA’ for LINE-1-rbz; red, average RA for LINE-1-ori.

This RA’ measure is calculated based on the cleaved sequences which were captured by RtcB ligation, while the RA used both cleaved and uncleaved sequences from the RNA-seq data.

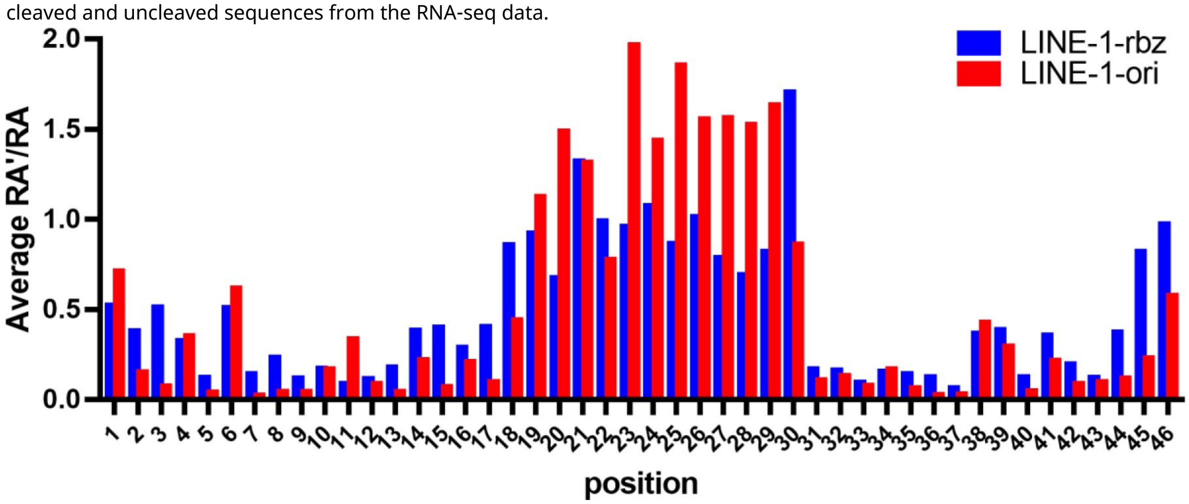


Figure S6.

(A) Structure descriptor for LINE-1-rbz. (B) Structure descriptor for OR4K15-rbz.

A

```
h1 s1 h2 s2 h2' s3 h1'
h1 0:1 NNNN:NNNN
s1 0 GCTCAA
h2 0:1 NNNNN:NNNNN
s2 0 N[99]
s3 0 GCHWG
```

B

```
h1 s1 h2 s2 h2' s3 h1'
h1 0:1 NNNNN:NNNNN
s1 0 GCHWG
h2 0:1 NNNN:NNNN
s2 0 N[99]
s3 0 GCTCAA
```

Figure S7.

(A) The structure of LINE-1-core in a surface representation, except for the C10- A11 site, which is shown in a stick representation. (B) Continuous stacking of stems P1 and P2. (C) Stacking interactions in loop L1 region.

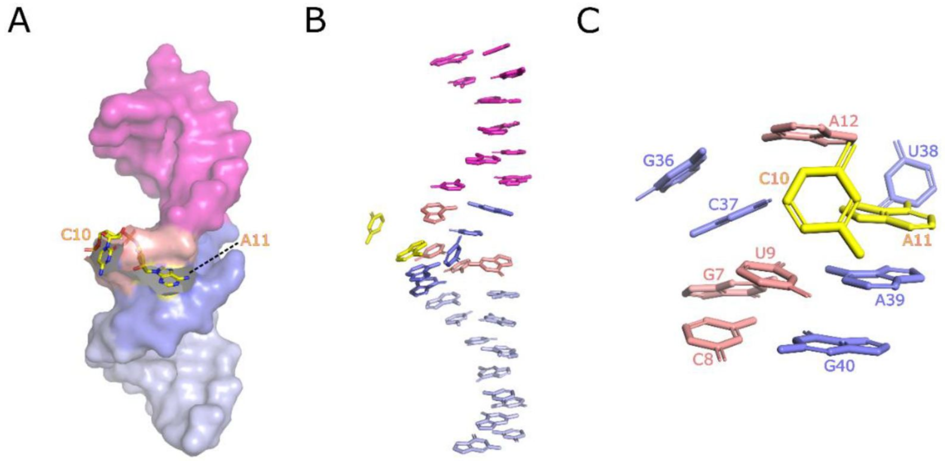


Figure S8.

Base-pairing interactions in loop L1 of LINE-1-core that extend the stem P1 and stem P2.

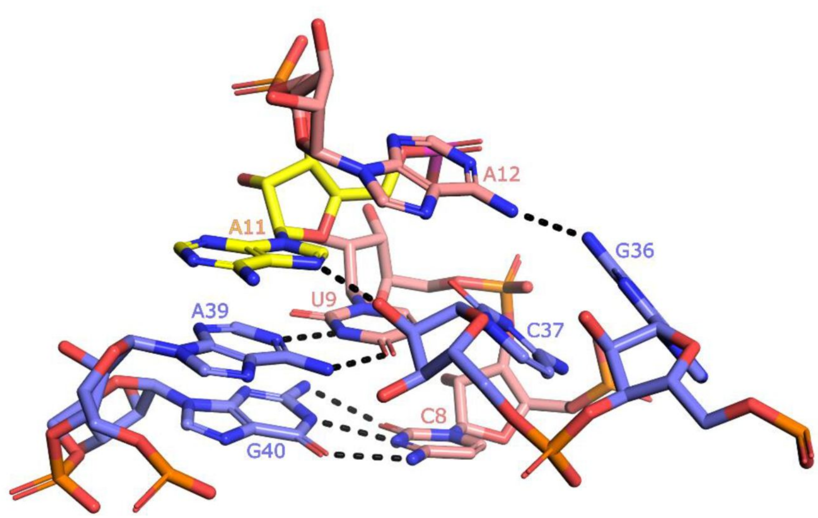


Figure S9.

(A) A detailed view of the interactions in loop L1 of OR4K15-core. (B) Stacking interactions in loop L1 region. (C) The network of H-bonding interactions involving G37. (D) Intermolecular contacts at the C40-A41 cleavage site in the modelled structure. The scissile phosphate is colored magenta.

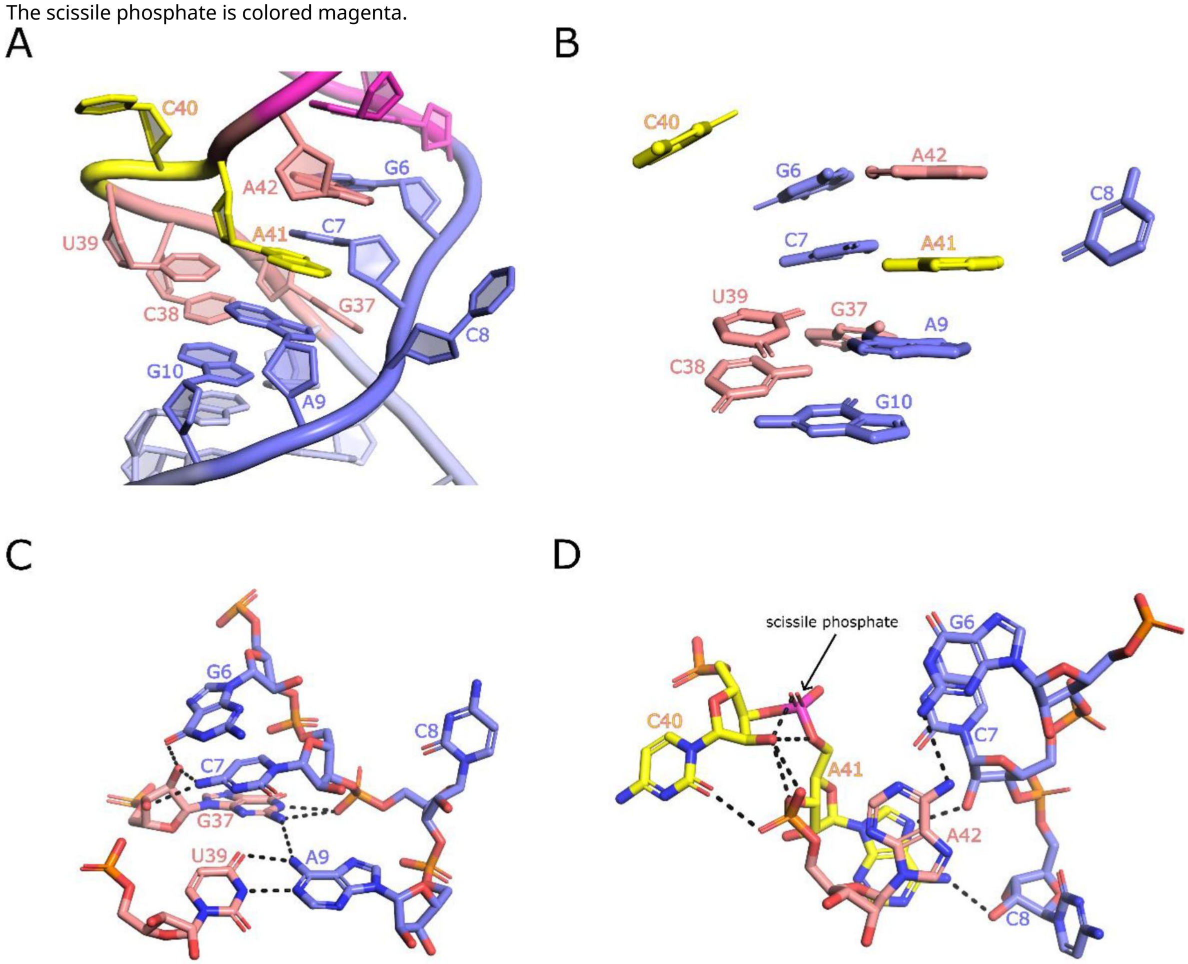


Figure S10.

PAGE-based cleavage assay of LINE-1-core in a 24-hour range.

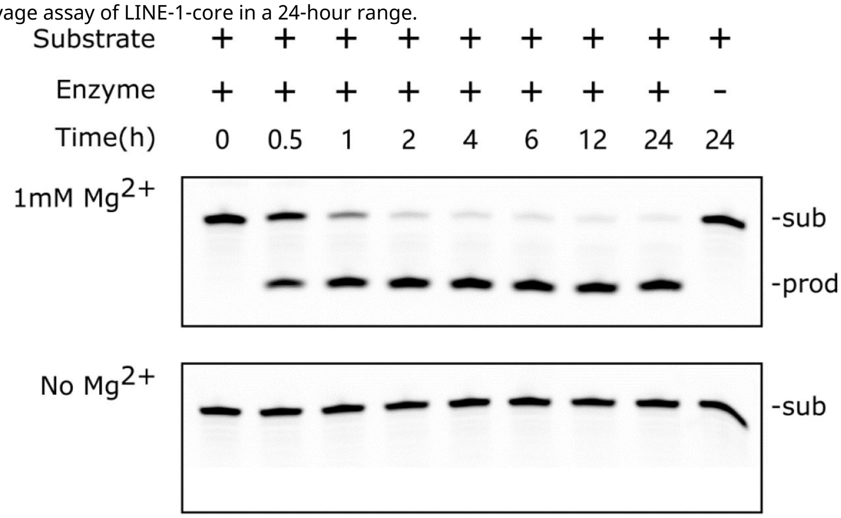


Figure S11.

The dependence of two lantern ribozymes' rate constants on Mg^{2+}/Mn^{2+} concentration.

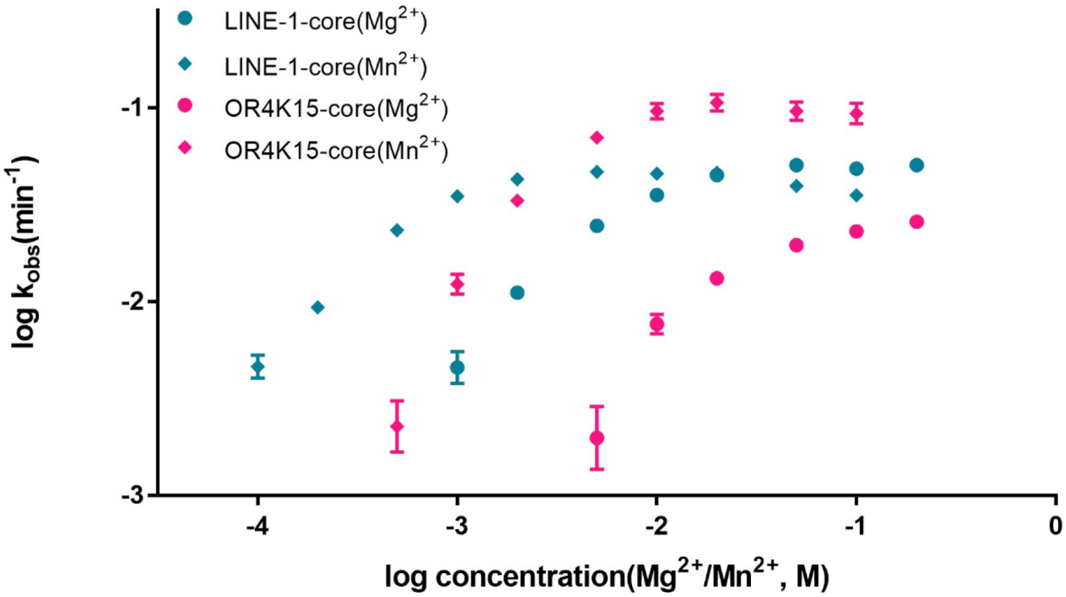


Figure S12.

A representative time course for the bimolecular ribozyme construct LINE-1- core under the conditions indicated as used for determining kobs values.

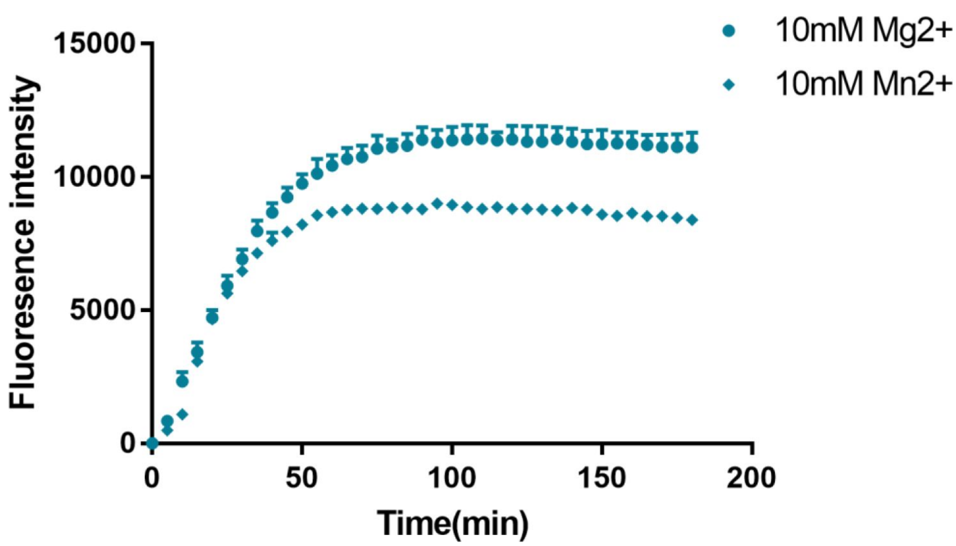


Figure S13.

The bimolecular ribozyme construct LINE-1-core with different E/S ratios (molar ratio between the enzyme strand and the substrate strand) for determining kobs values.

The final concentrations for the reaction solution were 20 mM Tris-HCl buffer (pH 7.5 at 23°C), 100 mM KCl, and 10 mM MgCl2. (A) An example of the time course assay for ribozyme cleavage. (B) The distribution of kobs values measured using different E/S ratios.

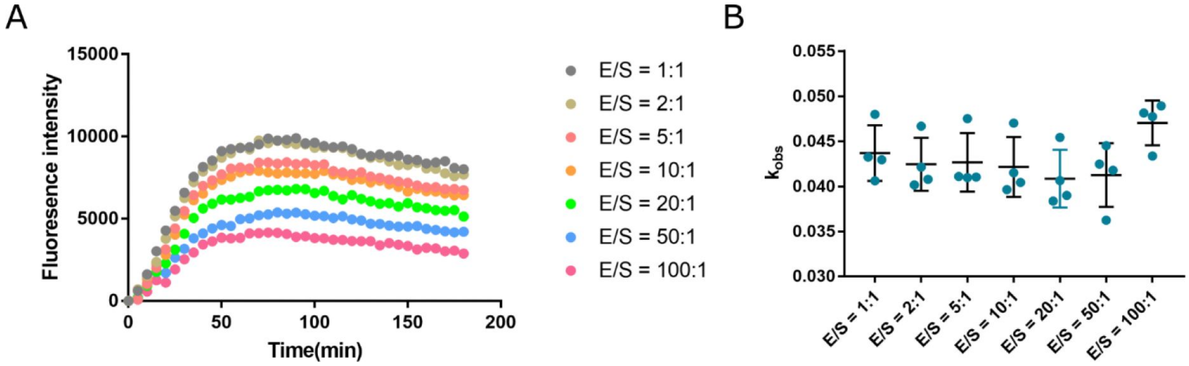
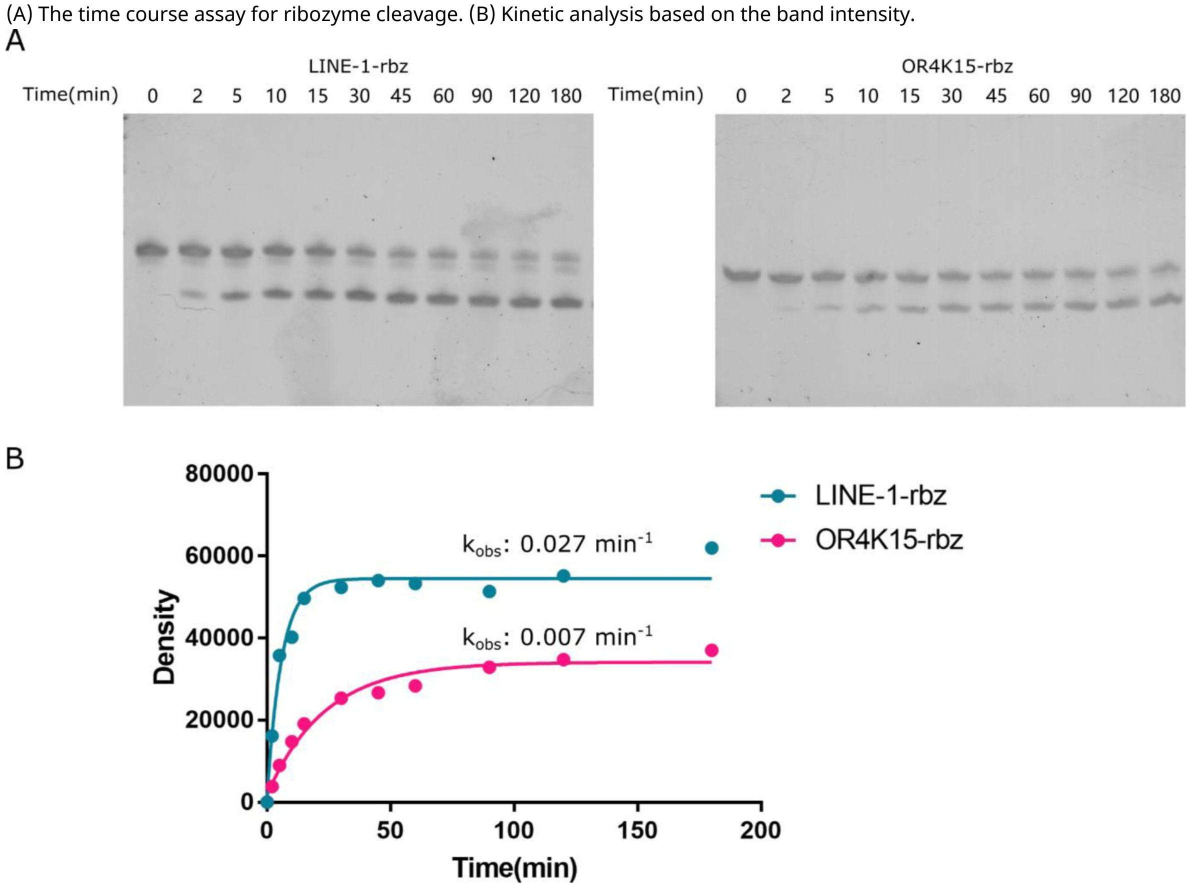


Figure S14.

Kinetic analysis using the single stranded LINE-1-rbz and OR4K15-rbz. The final concentrations for the reaction solution were 20 mM Tris-HCl buffer (pH 7.5 at 23°C), 100 mM KCl, and 10 mM MgCl2.

(A) The time course assay for ribozyme cleavage. (B) Kinetic analysis based on the band intensity.



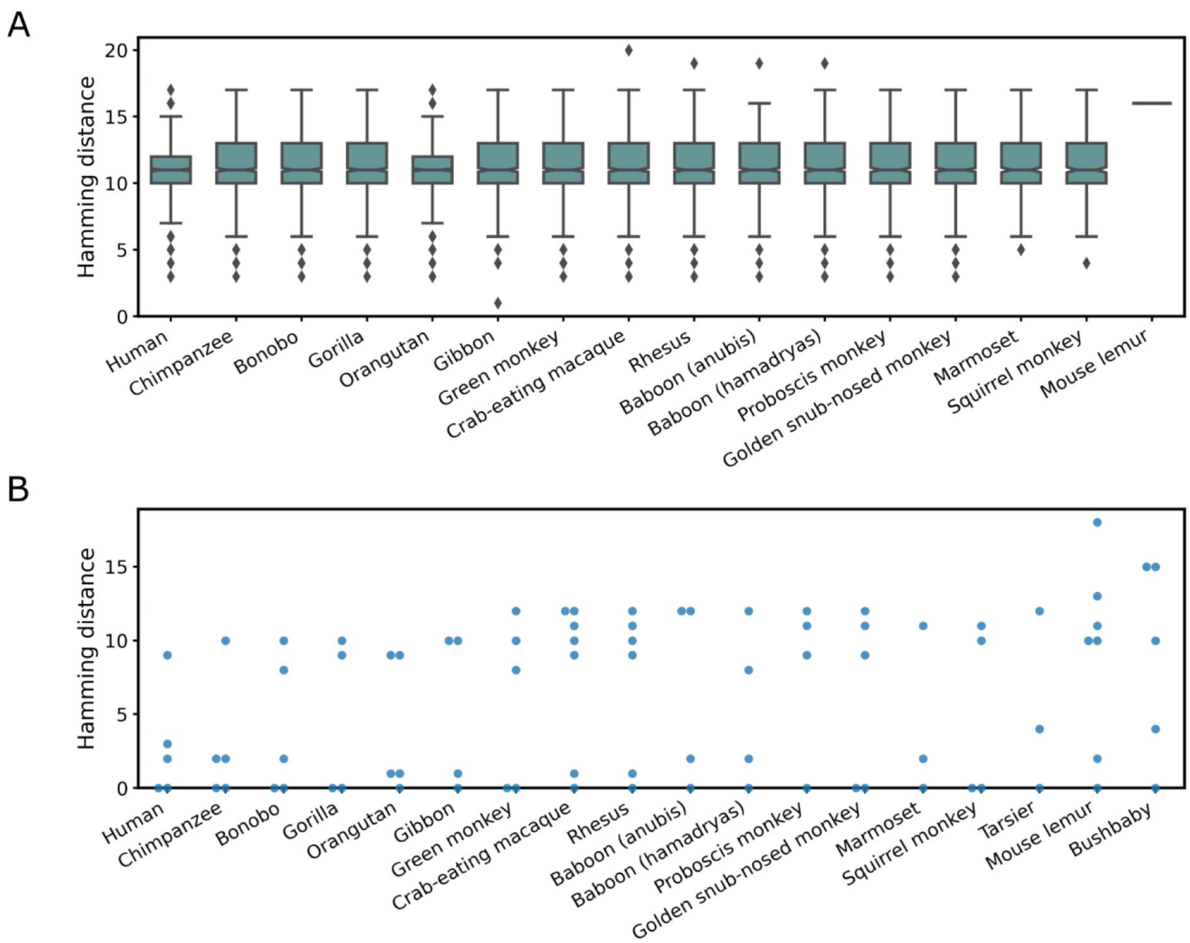


Figure S15.

(A) Hamming distance distribution of LINE-1-rbz homologs found in representative primate genome assemblies. (B) Hamming distance distribution of OR4K15- rbz homologs found in representative primate genome assemblies.

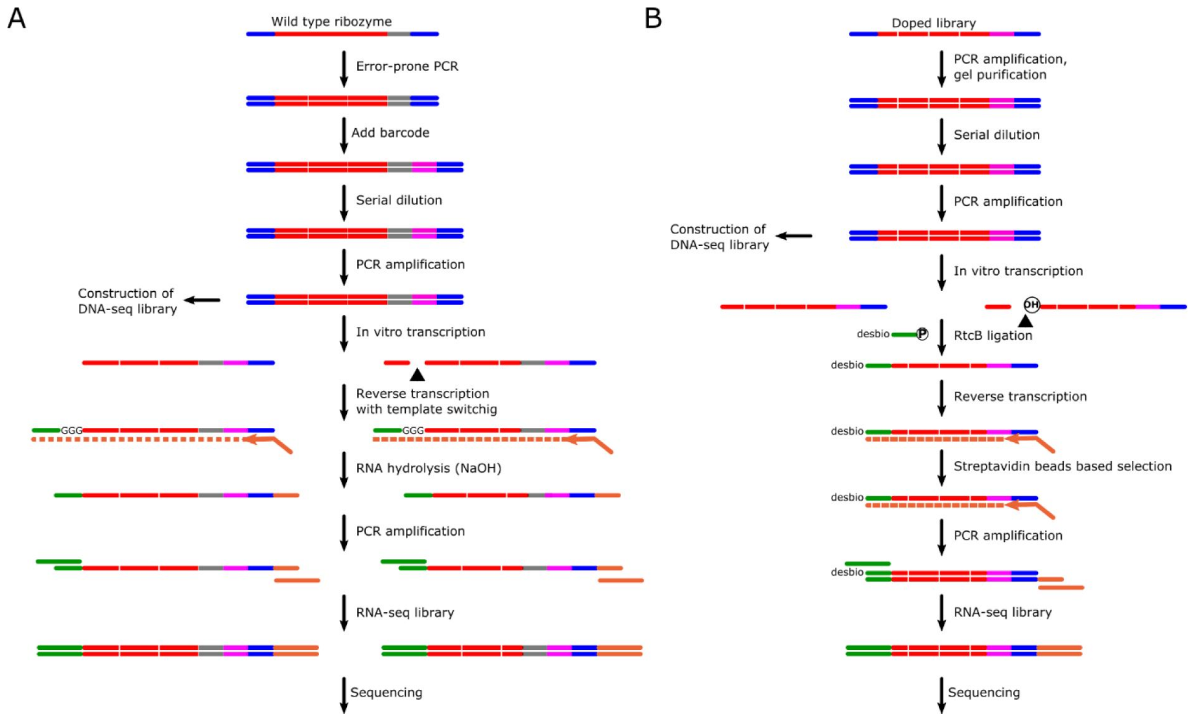


Figure S16.

Experimental pipeline of two deep mutational scanning experiments.

Red, ribozyme sequence; blue, T7 promoter and M13F sequence; grey, linker sequence; magenta, barcode sequence; green, template switching oligo and sequencing primer; orange, reverse transcription primer and sequencing primer. (A) The mutation library of the original sequence of LINE-1 ribozyme was generated by error-prone PCR. The RNA-seq library was constructed by using the cDNA synthesized by reverse transcription and template switching. (B) The chemically synthesized doped library was used to construct the mutation library of the contiguous, functional region of LINE-1 ribozyme. RtcB ligase was used to capture 3’ fragment after cleavage, then the ligated product was used to construct the RNA-seq library.

References

1. Benner S.A., Ellington A.D., Tauer A (1989) **Modem metabolism as a palimpsest of the RNA world** *Proc. Natl. Acad. Sci. U. S. A* **86**:7054–7058

2. Ferré-D'Amaré A.R., Scott W.G (2010) **Small Self-cleaving Ribozymes** *Cold Spring Harb. Perspect. Biol* **2**

3. Prody G.A., Bakos J.T., Buzayan J.M., Schneider I.R., Bruening G (1986) **Autolytic processing of dimeric plant vims satellite RNA** *Science* **231**:1577–1580

4. Forster A.C., Symons R.H (1987) **Self-cleavage of virusoid RNA is performed by the proposed 55-nucleotide active site** *Cell* **50**:9–16

5. Hutchins C.J., Rathjen P.D., Forster A.C., Symons R.H (1986) **Self-cleavage of plus and minus RNA transcripts of avocado sunblotch viroid** *Nucleic Acids Res* **14**:3627–3640

6. Cervera A., de la Peña M (2020) **Small circRNAs with self-cleaving ribozymes are highly expressed in diverse metazoan transcriptomes** *Nucleic Acids Res* **48**:5054–5064

7. de la Peña M., Garcia-Robles I (2010) **Intronic hammerhead ribozymes are ultraconserved in the human genome** *EMBO Rep* **11**:711–716

8. Salehi-Ashtiani K., Lupták A., Litovchick A., Szostak J.W (2006) **A genomewide search for ribozymes reveals an HDV-like sequence in the human CPEB3 gene** *Science* **313**:1788–1792

9. Martick M., Horan L.H., Noller H.F., Scott W.G (2008) **A discontinuous hammerhead ribozyme embedded in a mammalian messenger RNA** *Nature* **454**:899–902

10. Eickbush D.G., Eickbush T.H (2010) **R2 Retrotransposons Encode a Self-Cleaving Ribozyme for Processing from an rRNA Cotranscript** *Mol. Cell. Biol* **30**:3142–3150

11. Sánchez-Luque F.J., López M.C., Macias F., Alonso C., Thomas M.C (2011) **Identification of an hepatitis delta virus-like ribozyme at the mRNA 5′-end of the LITc retrotransposon from Trypanosoma cruzi** *Nucleic Acids Res* **39**:8065–8077

12. Perreault J., Weinberg Z., Roth A., Popescu O., Chartrand P., Ferbeyre G., Breaker R.R (2011) **Identification of Hammerhead Ribozymes in All Domains of Life Reveals Novel Structural Variations** *PLoS Comput. Biol* **7**

13. Chen Y., Qi F., Gao F., Cao H., Xu D., Salehi-Ashtiani K., Kapranov P (2021) **Hovlinc is a recently evolved class of ribozyme found in human IncRNA** *Nat. Chem. Biol* https://doi.org/10.1038/s41589-021-00763-0

14. Vogler C., Spalek K., Aemi A., Demougin P., Müller A., Huynh K.-D., Papassotiropoulos A., de Quervain D.J.-F (2009) **CPEB3 is Associated with Human Episodic Memory** *Front. Behav. Neurosci* **3**

15. Chen C.C. *et al.* (2023) **Inhibition of CPEB3 ribozyme elevates CPEB3 protein expression and polyadenylation of its target mRNAs, and enhances object location memory** *bioRxiv* https://doi.org/10.1101/2023.06.07.543953

16. Bendixsen D.P., Pollock T.B., Peri G., Hayden E.J (2021) **Experimental Resurrection of Ancestral Mammalian CPEB3 Ribozymes Reveals Deep Functional Conservation** *Mol. Biol. Evol* **38**:2843–2853

17. Ferbeyre G., Smith J.M., Cedergren R (1998) **Schistosome satellite DNA encodes active hammerhead ribozymes** *Mol. Cell. Biol* **18**:3880–3888

18. Cervera A., De la Pena M (2014) **Eukaryotic penelope-like retroelements encode hammerhead ribozyme motifs** *Mol. Biol. Evol* **31**:2941–2947

19. Liinse C.E., Weinberg Z., Breaker R.R (2017) **Numerous small hammerhead ribozyme variants associated with Penelope-like retrotransposons cleave RNA as dimers** *RNA Biol* **14**:1499–1507

20. Cervera A., Urbina D., de la Pena M (2016) **Retrozymes are a unique family of non-autonomous retrotransposons with hammerhead ribozymes that propagate in plants through circular RNAs** *Genome Biol* **17**

21. Dawid LB., Rebbert M.L (1981) **Nucleotide sequences at the boundaries between gene and insertion regions in the rDNA of Drosophilia melanogaster** *Nucleic Acids Res* **9**:5011–5020

22. Roiha H., Miller J.R., Woods L.C., Glover D.M (1981) **Arrangements and rearrangements of sequences flanking the two types of rDNA insertion in D. melanogaster** *Nature* **290**:749–753

23. Eickbush T.H., Robins B (1985) **Bombyx mori 28S ribosomal genes contain insertion elements similar to the Type I and II elements of Drosophila melanogaster** *EMBO J* **4**:2281–2285

24. Ruminski D.J., Webb C.-H.T., Riccitelli N.J., Luptak A (2011) **Processing and translation initiation of non-long terminal repeat retrotransposons by hepatitis delta virus (HDV)-like self-cleaving ribozymes** *J. Biol. Chern* **286**:41286–41295

25. Webb C.-H.T., Riccitelli N.J., Ruminski D.J., Luptäk A (2009) **Widespread occurrence of self-cleaving ribozymes** *Science* **326**

26. Zhang Z., Xiong P., Zhang T., Wang J., Zhan J., Zhou Y (2020) **Accurate inference of the full base-pairing structure of RNA by deep mutational scanning and covariation-induced deviation of activity** *Nucleic Acids Res* **48**:1451–1465

27. Cadwell R.C., Joyce G.F (1992) **Randomization of genes by PCR mutagenesis** *PCR Methods AppL* **2**:28–33

28. Keohavong P., Thilly W.G (1989) **Fidelity of DNA polymerases in DNA amplification** *Proc. Natl. Acad. Sci. U. S. A* **86**:9253–9257

29. Tanaka N., Chakravarty A.K., Maughan B., Shuman S (2011) **Novel mechanism of RNA repair by RtcB via sequential 2′,3′-cyclic phosphodiesterase and 3′- Phosphate/5′-hydroxyl ligation reactions** *J. Biol. Chem* **286**:43134–43143

30. Lemieux S., Major F (2002) **RNA canonical and non-canonical base pairing types: a recognition method and complete repertoire** *Nucleic Acids Res* **30**:4250–4263

31. Weinberg Z., Breaker R.R (2011) **R2R—software to speed the depiction of aesthetic consensus RNA secondary structures** *BMC Bioinformatics* **12**

32. Zheng L., Mairhofer E., Teplova M., Zhang Y., Ma J., Patel D.J., Micura R., Ren A (2017) **Structure-based insights into self-cleavage by a four-way junctional twistersister ribozyme** *Nat. Commun* **8**

33. Liu Y., Wilson T.J., Lilley D.M.J (2017) **The structure of a nucleolytic ribozyme that employs a catalytic metal ion** *Nat. Chem. Biol* **13**:508–513

34. Roth A., Weinberg Z., Chen A.G.Y., Kim P.B., Ames T.D., Breaker R.R (2014) **A widespread self-cleaving ribozyme class is revealed by bioinformatics** *Nat. Chem. Biol* **10**:56–60

35. Harris K.A., Liinse C.E., Li S., Brewer K.I., Breaker R.R (2015) **Biochemical analysis of pistol self-cleaving ribozymes** *RNA* **21**:1852–1858

36. Li S., Liinse C.E., Harris K.A., Breaker R.R (2015) **Biochemical analysis of hatchet self-cleaving ribozymes** *RNA* **21**:1845–1851

37. Weinberg Z., Kim P.B., Chen T.H., Li S., Harris K.A., Liinse C.E., Breaker R.R (2015) **New classes of self-cleaving ribozymes revealed by comparative genomics analysis** *Nat. Chem. Biol* **11**:606–610

38. Eddy S.R., Durbin R (1994) **RNA sequence analysis using covariance models** *Nucleic Acids Res* **22**:2079–2088

39. Deng J. *et al.* (2022) **Ribocentre: a database of ribozymes** *Nucleic Acids Res* **51**:D262–D268

40. Eickbush T.H., Eickbush D.G (2015) **Integration, regulation, and long-term stability of R2 retrotransposons** *Microbiol. Spectr* **3**

41. Lander E.S. *et al.* (2001) **Initial sequencing and analysis of the human genome** *Nature* **409**:860–921

42. Brouha B., Schustak J., Badge R.M., Lutz-Prigge S., Farley A.H., Moran J.V., Kazazian H.H (2003) **Hot L1s account for the bulk of retrotransposition in the human population** *Proc. Natl. Acad. Sci. U. S. A* **100**:5280–5285

43. Vivancos A.P., Gilell M., Dohm J.C., Serrano L., Himmelbauer H (2010) **Strandspecific deep sequencing of the transcriptome** *Genome Res* **20**:989–999

44. Seehafer C., Kalweit A., Steger G., Graf S., Hammann C (2011) **From alpaca to zebrafish: hammerhead ribozymes wherever you look** *RNA N. Y. N* **17**:21–26

45. Przybilski R., Graf S., Lescoute A., Nellen W., Westhof E., Steger G., Hammann C (2005) **Functional hammerhead ribozymes naturally encoded in the genome of Arabidopsis thaliana** *Plant Cell* **17**:1877–1885

46. de la Pena M., Garcia-Robles I (2010) **Ubiquitous presence of the hammerhead ribozyme motif along the tree of life** *RNA N. Y. N* **16**:1943–1950

47. Jimenez R.M., Delwart E., Luptak A (2011) **Structure-based Search Reveals Hammerhead Ribozymes in the Human Microbiomef** *J. Biol. Chem* **286**:7737–7743

48. Weinberg C.E., Olzog V.J., Eckert I., Weinberg Z (2021) **Identification of over 200- fold more hairpin ribozymes than previously known in diverse circular RNAs** *Nucleic Acids Res* **49**:6375–6388

49. De la Pena M., Gago S., Flores R (2003) **Peripheral regions of natural hammerhead ribozymes greatly increase their self-cleavage activity** *EMBO J* **22**:5561–5570

50. Zhao Z.Y., Wilson T.J., Maxwell K., Lilley D.M (2000) **The folding of the hairpin ribozyme: dependence on the loops and the junction** *RNA* **6**:1833–1846

51. Tinsley R.A., Walter N.G (2007) **Long-range impact of peripheral joining elements on structure and function of the hepatitis delta virus ribozyme** *Biol. Chem* **388**:705–715

52. Webb C.-H.T., Nguyen D., Myszka M., Luptak A (2016) **Topological constraints of structural elements in regulation of catalytic activity in HDV-like self-cleaving ribozymes** *Sci. Rep* **6**

53. Epstein L.M., Gall J.G (1987) **Self-cleaving transcripts of satellite DNA from the newt** *Cell* **48**:535–543

54. Riccitelli N.J., Delwart E., Luptak A (2014) **Identification of minimal HDV-like ribozymes with unique divalent metal ion dependence in the human microbiome** *Biochemistry* **53**:1616–1626

55. Huang X. *et al.* (2019) **Intracellular selection of trans-cleaving hammerhead ribozymes** *Nucleic Acids Res* **47**:2514–2522

56. Kobori S., Yokobayashi Y (2016) **High-Throughput Mutational Analysis of a Twister Ribozyme** *Angew. Chem. Int. Ed Engl* **55**:10354–10357

57. Zhang J., Kobert K., Flouri T., Stamatakis A (2014) **PEAR: a fast and accurate Illumina Paired-End reAd mergeR** *Bioinforma. Oxf. Engl* **30**:614–620

58. Ovchinnikov S., Park H., Varghese N., Huang P.-S., Pavlopoulos G.A., Kim D.E., Kamisetty H., Kyrpides N.C., Baker D (2017) **Protein structure determination using metagenome sequence data** *Science* **355**:294–298

59. Xiong P., Wu R., Zhan J., Zhou Y (2021) **Pairing a high-resolution statistical potential with a nucleobase-centric sampling algorithm for improving RNA model refinement** *Nat. Commun* **12**

60. Nawrocki E.P., Kolbe D.L., Eddy S.R (2009) **Infernal 1.0: inference of RNA alignments** *Bioinforma. Oxf. Engl* **25**:1335–1337

61. Nawrocki E.P., Eddy S.R (2013) **Infernal 1.1: 100-fold faster RNA homology searches** *Bioinformatics* **29**:2933–2935

Editors

Reviewing Editor
Aaron Frank
Arrakis Therapeutics, Waltham, United States of America

Senior Editor
Amy Andreotti
Iowa State University, Ames, United States of America

Reviewer #1 (Public review):

Summary:

The overall analysis and discovery of the common motif is important and exciting. Very few human/primate ribozymes have been published and this manuscript presents a detailed analysis of two of them. The minimized domains appear to be some of the smallest known self-cleaving ribozymes.

Strengths:

The manuscript is rooted in deep mutational analysis of the human OR4K15 and LINE1 ribozymes and subsequently in modeling of their active site based on the closely-related core of the TS ribozyme. The experiments support the HTS findings and provide convincing evidence that the ribozymes are structurally related to the core of the TS ribozyme, which has not been found in primates prior to this work.

https://doi.org/10.7554/eLife.90254.2.sa1

Author response:

The following is the authors' response to the original reviews.

> ***Public Reviews:***
>
> ***Reviewer #1 (Public Review):***
>
> *Summary:*
>
> *The overall analysis and discovery of the common motif are important and exciting. Very few human/primate ribozymes have been published and this manuscript presents a relatively detailed analysis of two of them. The minimized domains appear to be some of the smallest known self-cleaving ribozymes.*
>
> *Strengths:*
>
> *The manuscript is rooted in deep mutational analysis of the OR4K15 and LINE1 and subsequently in modeling of a huge active site based on the closely-related core of the TS ribozyme. The experiments support the HTS findings and provide convincing evidence that the ribozymes are structurally related to the core of the TS ribozyme, which has not been found in primates prior to this work.*
>
> *Weaknesses:*
>
> *(1) Given that these two ribozymes have not been described outside of a single figure in a Science Supplement, it is important to show their locations in the human genome,*

> *present their sequence and structure conservation among various species, particularly primates, and test and discuss the activity of variants found in non-human organisms. Furthermore, OR4K15 exists in three copies on three separate chromosomes in the human genome, with slight variations in the ribozyme sequence. All three of these variants should be tested experimentally and their activity should be presented. A similar analysis should be presented for the naturally-occurring variants of the LINE1 ribozyme. These data are a rich source for comparison with the deep mutagenesis presented here. Inserting a figure (1) that would show the genomic locations, directions, and conservation of these ribozymes and discussing them in light of this new presentation would greatly improve the manuscript. As for the biological roles of known self-cleaving ribozymes in humans, there is a bioRxiv manuscript on the role of the CPEB3 ribozyme in mammalian memory formation (doi.org/10.1101/2023.06.07.543953), and an analysis of the CPEB3 functional conservation throughout mammals (Bendixsen et al. MBE 2021). Furthermore, the authors missed two papers that presented the discovery of human hammerhead ribozymes that reside in introns (by de la PeÃ{plus minus}a and Breaker), which should also be cited. On the other hand, the Clec ribozyme was only found in rodents and not primates and is thus not a human ribozyme and should be noted as such.*

We thank this Reviewer for his/her input and acknowledgment of this work. To improve the manuscript, we have included the genomic locations in Figure 1A, Figure 6A and Figure 6C. And we have tested the activity of representative variants found in the human genome and discussed the activity of the variants in other primates. All suggested publications are now properly cited.

Line 62-66: It has been shown that single nucleotide polymorphism (SNP) in CPEB3 ribozyme was associated with an enhanced self-cleavage activity along with a poorer episodic memory (14). Inhibition of the highly conserved CPEB3 ribozyme could strengthen hippocampal-dependent long-term memory (15, 16). However, little is known about the other human self-cleaving ribozymes.

Line 474-501: Homology search of two TS-like ribozymes. To locate close homologs of the two TS-like ribozymes, we performed cmsearch based on a covariance model (38) built on the sequence and secondary structural profiles. In the human genome, we got 1154 and 4 homolog sequences for LINE-1-rbz and OR4K15-rbz, respectively. For OR4K15-rbz, there was an exact match located at the reverse strand of the exon of OR4K15 gene (Figure 6A). The other 3 homologs of OR4K15-rbz belongs to the same olfactory receptor family 4 subfamily K (Figure 6C). However, there was no exact match for LINE-1-rbz (Figure 6A). Interestingly, a total of 1154 LINE-1-rbz homologs were mapped to the LINE-1 retrotransposon according to the RepeatMasker (http://www.repeatmasker.org) annotation. Figure 6B showed the distribution of LINE-1-rbz homologs in different LINE-1 subfamilies in the human genome. Only three subfamilies L1PA7, L1PA8 and L1P3 (L1PA7-9) can be considered as abundant with LINE-1-rbz homologs (>100 homologs per family). The consensus sequences of all homologs obtained are shown in Figure 6D. In order to investigate the self-cleavage activity of these homologs, we mainly focused on the mismatches in the more conserved internal loops. The major differences between the 5 consensus sequences are the mismatches in the first internal loop. The widespread A12C substitution can be found in majority of LINE-1-rbz homologs, this substitution leads to a one-base pair extension of the second stem (P2) but almost no activity (RA': 0.03) based on our deep mutational scanning result. Then we selected 3 homologs without A12C substitution for LINE-1-rbz for in vitro cleavage assay (Figure 6E). But we didn't observe significant cleavage activity, this might be caused by GU substitutions in the stem region. For 3 homologs of OR4K15-rbz, we only found one homolog of OR4K15 with pronounced self-cleavage activity (Figure 6F). In addition, we performed similar bioinformatic search of the TS-like ribozymes in other primate genomes. Similarly, the majority (15 out of 18) of primate genomes have a large number of LINE-1 homologs (>500) and the remaining three have essentially none. However, there was no exact match. Only one homolog has a single mutation (U38C) in the genome assembly of Gibbon (Figure S15). The majority of these homologs have 3 or more mismatches (Figure S15). For OR4K15-rbz, all representative primate genomes contain at least one exact match of the OR4K15-rbz sequence.

Line 598-602: According to the bioinformatic analysis result, there are some TS-like ribozymes (one LINE-1-rbz homolog in the Gibbon genome, and some OR4K15-rbz homologs) with in vitro cleavage activity in primate genomes. Unlike the more conserved CPEB3 ribozyme which has a clear function, the function of the TS-like ribozymes is not clear, as they are not conserved, belong to the pseudogene or located at the reverse strand.

> *(2) The authors present the story as a discovery of a new RNA catalytic motif. This is unfounded. As the authors point out, the catalytic domain is very similar to the Twister Sister (or "TS") ribozyme. In fact, there is no appreciable difference between these and TS ribozymes, except for the missing peripheral domains. For example, the env33 sequence in the Weinberg et al. 2015 NCB paper shows the same sequences in the catalytic core as the LINE1 ribozyme, making the LINE1 ribozyme a TS-like ribozyme in every way, except for the missing peripheral domains. Thus these are not new ribozymes and should not have a new name. A more appropriate name should be TS-like or TS-min ribozymes. Renaming the ribozymes to lanterns is misleading.*

Although we observed some differences in mutational effects, we agree with the reviewer that it is more appropriate to call them TS-like ribozymes. We have replaced all “lantern ribozyme” with “TS-like ribozyme” as suggested.

> *(3) In light of 2) the story should be refocused on the fact the authors discovered that the OR4K15 and LINE1 are both TS-like ribozymes. That is very exciting and is the real contribution of this work to the field.*

We thank this Reviewer for their acknowledgement of this work. To improve the manuscript, we have re-named the ribozymes as suggested.

> *(4) Given the slow self-scission of the OR4K15 and LINE1 ribozymes, the discussion of the minimal domains should be focused on the role of peripheral domains in full-length TS ribozymes. Peripheral domains have been shown to greatly speed up hammerhead, HDV, and hairpin ribozymes. This is an opportunity to show that the TS ribozymes can do the same and the authors should discuss the contribution of peripheral domains to the ribozyme structure and activity. There is extensive literature on the contribution of a tertiary contact on the speed of self-scission in hammerhead ribozymes, in hairpin ribozyme it's centered on the 4-way junction vs 2-way junction structure, and in HDVs the contribution is through the stability of the J1/2 region, where the stability of the peripheral domain can be directly translated to the catalytic enhancement of the ribozymes.*

We appreciate your question and the valuable suggestions provided. We have included the citations and discussion about the peripheral domains in other ribozymes.

Line 570-576: Thus, a more sophisticated structure along with long-range interactions involving the SL4 region in the twister sister ribozyme must have helped to stabilize the catalytic region for the improved catalytic activity. Similarly, previous studies have demonstrated that peripheral regions of hammerhead (49), hairpin (50) and HDV (51, 52) ribozymes could greatly increase their self-cleavage activity. Given the importance of the peripheral regions, absence of this tertiary interaction in the TS-like ribozyme may not be able to fully stabilize the structural form generated from homology modelling.

> *(5) The argument that these are the smallest self-cleaving ribozymes is debatable. LÃ1/4nse et al (NAR 2017) found some very small hammerhead ribozymes that are smaller than those presented here, but the authors suggest only working as dimers. The human ribozymes described here should be analyzed for dimerization as well (e.g., by native gel analysis) particularly because the authors suggest that there are no peripheral domains that stabilize the fold. Furthermore, Riccitelli et al. (Biochemistry) minimized the HDV-like ribozymes and found some in metagenomic sequences that are about the same size as the ones presented here. Both of these papers should be cited and discussed.*

We apologize for any confusion caused by our previous statement. To clarify, we highlighted "35 and 31 nucleotides only" because 46 and 47 nt contain the variable hairpin loops which are not important for the catalytic activity. By comparing the conserved segments, the TS-like ribozyme discussed in this paper is the shortest with the simplest secondary structure. And we have replaced the terms "smallest" and "shortest" with "simplest" in our manuscript. The title has been changed to "Minimal twister sister (TS)-like self-cleaving ribozymes in the human genome revealed by deep mutational scanning". All the publications mentioned have been cited and discussed. Regarding possible dimerization, we did not find any evidence but would defer it to future detailed structural analysis to be sure.

Line 605-608: Previous studies also have revealed some minimized forms of self-cleaving ribozymes, including hammerhead (19, 53) and HDV-like (54) ribozymes. However, when comparing the conserved segments, they (>= 36 nt) are not as short as the TS-like ribozymes (31 nt) found here.

> *(6) The authors present homology modeling of the OR4K15 and LINE1 ribozymes based on the crystal structures of the TS ribozymes. This is another point that supports the fact that these are not new ribozyme motifs. Furthermore, the homology model should be carefully discussed as a model and not a structure. In many places in the text and the supplement, the models are presented as real structures. The wording should be changed to carefully state that these are models based on sequence similarity to TS ribozymes. Fig 3 would benefit from showing the corresponding structures of the TS ribozymes.*

We thank the reviewer for pointing these out and we have already fixed them. We have replaced all "lantern ribozyme" with "TS-like ribozyme" as suggested. The term "Modelled structures" were used for representing the homology model. And we have included the TS ribozyme structure in Fig 3.

> ***Reviewer #2 (Public Review):***
>
> *Summary:*
>
> *This manuscript applies a mutational scanning analysis to identify the secondary structure of two previously suggested self-cleaving ribozyme candidates in the human genome. Through this analysis, minimal structured and conserved regions with imminent importance for the ribozyme's activity are suggested and further biochemical evidence for cleavage activity are presented. Additionally, the study reveals a close resemblance of these human ribozyme candidates to the known self-cleaving ribozyme class of twister sister RNAs. Despite the high conservation of the catalytic core between these RNAs, it is suggested that the human ribozyme examples constitute a new ribozyme class. Evidence for this however is not conclusive.*
>
> *Strengths:*

> *The deep mutational scanning performed in this study allowed the elucidation of important regions within the proposed LINE-1 and OR4K15 ribozyme sequences. Part of the ribozyme sequences could be assigned a secondary structure supported by covariation and highly conserved nucleotides were uncovered. This enabled the identification of LINE-1 and OR4K15 core regions that are in essence identical to previously described twister sister self-cleaving RNAs.*
>
> *Weaknesses:*
>
> *I am skeptical of the claim that the described catalytic RNAs are indeed a new ribozyme class. The studied LINE-1 and OR4K15 ribozymes share striking features with the known twister sister ribozyme class (e.g. Figure 3A) and where there are differences they could be explained by having tested only a partial sequence of the full RNA motif. It appears plausible, that not the entire "functional region" was captured and experimentally assessed by the authors.*

We thank this Reviewer for his/her input and acknowledgment of this work. Because a similar question was raised by reviewer 1, we decided to name the ribozymes as TS-like ribozymes. Regarding the entire regions, we conducted mutational scanning experiments at the beginning of this study. The relative activity distributions (Figure 1B, 1C) have shown that only parts of the sequence contributes to the self-cleavage activity. That is the reason why we decided to focus on the parts of the sequence afterwards.

> *They identify three twister sister ribozymes by pattern-based similarity searches using RNA-Bob. Also comparing the consensus sequence of the relevant region in twister sister and the two ribozymes in this paper underlines the striking similarity between these RNAs. Given that the authors only assessed partial sequences of LINE-1 and OR4K15, I find it highly plausible that further accessory sequences have been missed that would clearly reveal that "lantern ribozymes" actually belong to the twister sister ribozyme class. This is also the reason I do not find the modeled structural data and biochemical data results convincing, as the differences observed could always be due to some accessory sequences and parts of the ribozyme structure that are missing.*

We appreciate the reviewer for raising this question. As we explained in the last question, we now called the ribozymes as TS-like ribozymes. We also emphasize that the relative activity data of the original sequences have indicated that the other part did not make any contribution to the activity of the ribozyme. The original sequences provided in the Science paper (Salehi-Ashtiani et al. Science 2006) were generated from biochemical selection of the genomic library. It did not investigate the contribution of each position to the self-cleavage activity.

> *Highly conserved nucleotides in the catalytic core, the need for direct contacts to divalent metal ions for catalysis, the preference of Mn2+ oder Mg2+ for cleavage, the plateau in observed rate constants at ~100mM Mg2+, are all characteristics that are identical between the proposed lantern ribozymes and the known twister sister class.*
>
> *The difference in cleavage speed between twister sister (~5 min-1) and proposed lantern ribozymes could be due to experimental set-up (true single-turnover kinetics?) or could be explained by testing LINE-1 or OR4K15 ribozymes without needed accessory sequences. In the case of the minimal hammerhead ribozyme, it has been previously observed that missing important tertiary contacts can lead to drastically reduced cleavage speeds.*

We thank the reviewer for this question. We now called the ribozymes as TS-like ribozymes. As we explained in the last question, the relative activity data of the original sequences have proven that the other part did not make any contribution to the activity of the ribozyme. Moreover, we have tested different enzyme to substrate ratios to achieve single turn-over kinetics (Figure S13). The difference in cleavage speed should be related to the absence of peripheral regions which do not exist in the original sequences of the LINE-1 and OR4K15 ribozyme. We have included the publications and discussion about the peripheral domains in other ribozymes.

Line 458-463: The kobs of LINE-1-core was ~0.05 min-1 when measured in 10mM MgCl2 and 100mM KCl at pH 7.5 (Figure S13). Furthermore, the single-stranded ribozymes exhibited lower kobs (~0.03 min-1 for LINE-1-rbz) (Figure S14) when comparing with the bimolecular constructs. This confirms that the stem loop region SL2 does not contribute much to the cleavage activity of the TS-like ribozymes.

Line 570-576: Thus, a more sophisticated structure along with long-range interactions involving the SL4 region in the twister sister ribozyme must have helped to stabilize the catalytic region for the improved catalytic activity. Similarly, previous studies have demonstrated that peripheral regions of hammerhead (49), hairpin (50) and HDV (51, 52) ribozymes could greatly increase their self-cleavage activity. Given the importance of the peripheral regions, absence of this tertiary interaction in the TS-like ribozyme may not be able to fully stabilize the structural form generated from homology modelling.

> ***Reviewer 2: (Recommendations For The Authors):***
>
> *Major points*
>
> *It would have made it easier to connect the comments to text passages if the submitted manuscript had page numbers or even line numbers.*

We thank the reviewer for pointing this out and we have already fixed it.

> *In the introduction: "...using the same technique, we located the functional and base-pairing regions of..." The use of the adjective functional is imprecise. Base-paired regions are also important for the function, so what type of region is meant here? Conserved nucleotides?*

We thank the reviewer for pointing this out. We were describing the regions which were essential for the ribozyme activity. And we have defined the use of "functional region" in introduction.

Line 95: we located the regions essential for the catalytic activities (the functional regions) of LINE-1 and OR4K15 ribozymes in their original sequences.

> *In their discussion, the authors mention the possible flaws in their 3D-modelling in the absence of Mg2+. Is it possible to include this divalent metal ion in the calculations?*

We thank the reviewer for this question. Currently, BriQ (Xiong et al. Nature Communications 2021) we used for modeling doesn't include divalent metal ion in modeling.

Xiong, Peng, Ruibo Wu, Jian Zhan, and Yaoqi Zhou. 2021. "Pairing a High-Resolution Statistical Potential with a Nucleobase-Centric Sampling Algorithm for Improving RNA Model Refinement." Nature Communications 12: 2777. doi:10.1038/s41467-021-23100-4.

> *Abstract:*
>
> *It is claimed that ribozyme regions of 46 and 47 nt described in the manuscript resemble the shortest known self-cleaving ribozymes. This is not correct. In 1988, hammerhead*

> *ribozymes in newts were first discovered that are only 40 nt long.*

We apologize for any confusion caused by our previous statement. To clarify, we highlighted “35 and 31 nucleotides only” as 46 and 47 nt contain the variable hairpin loops which are not important for the catalytic activity. By comparing the conserved segments, the TS-like ribozyme discussed in this paper is the shortest with the simplest secondary structure. And we have replaced the terms “smallest” and “shortest” with “simplest” in our manuscript. The title has been changed to “Minimal TS-like self-cleaving ribozyme revealed by deep mutational scanning”.

> *The term "functional region" is, to my knowledge, not a set term when discussing ribozymes. Does it refer to the catalytic core, the cleavage site, the acid and base involved in cleavage, or all, or something else? Therefore, the term should be 1) defined upon its first use in the manuscript and 2) probably not be used in the abstract to avoid confusion to the reader.*

We apologize for any confusion caused by our previous statement. To clarify, we have changed the term “functional region” in abstract. And we have defined the use of “functional region” in introduction.

Line 34-37: We found that the regions essential for ribozyme activities are made of two short segments, with a total of 35 and 31 nucleotides only. The discovery makes them the simplest known self-cleaving ribozymes. Moreover, the essential regions are circular permutated with two nearly identical catalytic internal loops, supported by two stems of different lengths.

Line 95: we located the regions essential for the catalytic activities (the functional regions) of LINE-1 and OR4K15 ribozymes in their original sequences.

> *The choice of the term "non-functional loop" in the abstract is a bit unfortunate. The loop might not be important for promoting ribozyme catalysis by directly providing, e.g. the acid or base, but it has important structural functions in the natural RNA as part of a hairpin structure.*

We thank the reviewer for pointing this out and we have re-phrased the sentences.

Line 33-34: We found that the regions essential for ribozyme activities are made of two short segments, with a total of 35 and 31 nucleotides only.

Line 283: Removing the peripheral loop regions (Figures 1B and 1C) allows us to recognize that the secondary structure of OR4K15-rbz is a circular permutated version of LINE-1-rbz.

> *Results:*
>
> *Please briefly explain CODA and MC analysis when first mentioned in the results (Figure (1) The more detailed explanation of these terms for Figure 2 could be moved to this part of the results section (including explanations in the figure legend).*

We thank the reviewer for pointing this out and we included a brief explanation.

Line 150-154: CODA employed Support Vector Regression (SVR) to establish an independent-mutation model and a naive Bayes classifier to separate bases paired from unpaired (26). Moreover, incorporating Monte-Carlo simulated annealing with an energy model and a CODA scoring term (CODA+MC) could further improve the coverage of the regions under-sampled by deep mutations.

> *Please indicate the source of the human genomic DNA. Is it a patient sample, what type of tissue, or is it an immortalized cell line? It is not stated in the methods I believe.*

We thank the reviewer for pointing this out. According to the original Science paper (Salehi-Ashtiani et al. Science 2006), the human genomic DNA (isolated from whole blood) was purchased from Clontech (Cat. 6550-1). In our study, we directly employed the sequences provided in Figure S2 of the Science paper for gene synthesis. Thus, we think it is unnecessary to mention the source of genomic DNA in the methods section of our paper.

> *Please also refer to the methods section when the calculation of RA and RA' values is explained in the main text to avoid confusion.*

We thank the reviewer for pointing this out and we have fixed it.

Line 207-208: Figure 2A shows the distribution of relative activity (RA', measured in the second round of mutational scanning) (See Methods) of all single mutations

> *For OR4K15 it is stated that the deep mutational scanning only revealed two short regions as important. However, there is another region between approx. 124-131 nt and possibly even at positions 47 and 52 (to ~55), that could contribute to effective RNA cleavage, especially given the library design flaws (see below) and the lower mutational coverage for OR4K15. A possible correlation of the mutations in these regions is even visible in the CODA+MC analysis shown in Figure 1D on the left. Why are these regions ignored in ongoing experiments?*

We thank the reviewer for this question. As shown in Table S1, although the double mutation coverage of OR4K15-ori was low (16.2 %), we got 97.6 % coverage of single mutations. The relative activity of these single mutations was enough to identify the conserved regions in this ribozyme. Mutations at the positions mentioned by the reviewer did not lead to large reductions in relative activity. Since the relative activity of the original sequence is 1, we presumed that only positions with average relative activity much lower than 1 might contribute to effective cleavage.

Regarding the corresponding correlation of mutations in CODA+MC, they are considered as false positives generated from Monte Carlo simulated annealing (MC), because lack of support from the relative activity results.

> *Have the authors performed experiments with their "functional regions" in comparison to the full-length RNA or partial truncations of the full-length RNA that included, in the case of OR4K15, nt 47-131? Also for LINE-1 another stem region was mentioned (positions 14-18 with 30-34) and two additional base pairs. Were they included in experiments not shown as part of this manuscript?*

We appreciate the reviewer for raising this question. We only compared the full-length or partial truncations of the LINE-1 ribozyme. Since the secondary structure predicted from OR4K15-ori data was almost the same as LINE-1, we didn't perform deep mutagenesis on the partial truncation of the OR4K15. However, the secondary structure of OR4K15 was confirmed by further biochemical experiments.

Regarding the second question, the additional base pairs were generated by Monte Carlo simulated annealing (MC). They are considered as false positives because of low probabilities and lack of support from the deep mutational scanning results. The appearance of false positives is likely due to the imperfection of the experiment-based energy function employed in current MC simulated annealing.

> *Are there other examples in the literature, where error-prone PCR generates biases towards A/T nucleotides as observed here? Please cite!*

We thank the reviewer for pointing this out and we have included the corresponding citation.

Line 161-162: The low mutation coverage for OR4K15-ori was due to the mutational bias (27, 28) of error-prone PCR (Supplementary Figures S1, S2, S3 and S4).

Line 170-171: whose covariations are difficult to capture by error-prone PCR because of mutational biases (27, 28).

> *The authors mention that their CODA analysis was based on the relative activities of 45,925 and 72,875 mutation variants. I cannot find these numbers in the supplementary tables. They are far fewer than the read numbers mentioned in Supplementary Table 2. How do these numbers (45,925 and 72,875) arise? Could the authors please briefly explain their selection process?*

We apologize for any confusion caused by our previous statement. Our CODA analysis only utilized variants with no more than 3 mutations. The number listed in the supplementary tables is the total number of the variants. To clarify, we have included a brief explanation for these numbers.

Line 203-204: We performed the CODA analysis (26) based on the relative activities of 45,925 and 72,875 mutation variants (no more than 3 mutations) obtained for the original sequence and functional region of the LINE-1 ribozyme, respectively.

> *What are the reasons the authors assume their findings from LINE-1 can be used to directly infer the structure for OR4K15? (Third section in results, last paragraph)*

We apologize for any confusion caused by our previous statement. We meant to say that the consistency between LINE-1-rbz and LINE-1-ori results suggested that our method for inferring ribozyme structure was reliable. Thus, we employed the same method to infer the structure of the functional region of OR4K15. To clarify, we have re-phrased the sentence.

Line 259-261: The consistent result between LINE-1-rbz and LINE-1-ori suggested that reliable ribozyme structures could be inferred by deep mutational scanning. This allowed us to use OR4K15-ori to directly infer the final inferred secondary structure for the functional region of OR4K15.

> *There are several occasions where the authors use the differences between the proposed lantern ribozymes and twister sister data as reasons to declare LINE-1 and OR4K15 a new ribozyme class. As mentioned previously, I am not convinced these differences in structure and biochemical results could not simply result from testing incomplete LINE-1 and OR4K15 sequences.*

We apologize for any confusion caused by our previous statement. Despite we observed some differences in mutational effects, we agree with the reviewer that it is not convincing to claim them as a new ribozyme class. We have replaced all “lantern ribozyme” with “TS-like ribozyme” as the reviewer 1 suggested.

> *The authors state, that "the result confirmed that the stem loop SL2 region in LINE-1 and OR4K15 did not participate in the catalytic activity". To draw such a conclusion a kinetic comparison between a construct that contains SL2 and does not contain SL2 would be necessary. The given data does not suffice to come to this conclusion.*

We appreciate the reviewer for raising this question. To address this, we performed gel-based kinetic analysis of these two ribozymes (Figure S14).

Line 458-462: The kobs of LINE-1-core under single-turnover condition was ~0.05 min-1 when measured in 10mM MgCl2 and 100mM KCl at pH 7.5 (Figure S13). Only a slightly lower value of kobs (~0.03 min-1) was observed for LINE-1-rbz (Figure S14). This confirms that the stem loop region SL2 does not contribute to the cleavage activity of the TS-like ribozymes.

> *Construct/Library design:*
>
> *The last 31 bp in the OR4K15 ribozyme template sequence are duplicated (Supplementary Table 4). Therefore, there are 2 M13 fwd binding sites and several possible primer annealing sites present in this template. This could explain the lower yield for the mutational analysis experiments. Did the authors observe double bands in their PCR and subsequent analysis? The experiments should probably be repeated with a template that does not contain this duplication. Alternatively, the authors should explain, why this template design was chosen for OR4K15.*

We apologize for this mistake during writing. Our construct design for OR4K15 contains only one M13F binding site. We thank the reviewer for pointing this out and we have fixed the error.

> *Figure 5B: Where are the bands for the OR4K15 dC-substrate? They are not visible on the gel, so one has to assume there was no substrate added, although the legend indicates otherwise.*
>
> *Also this figure, please indicate here or in the methods section what kind of marker was used. In panels A and B, please label the marker lanes.*

We apologize for this mistake and we have repeated the experiment. The marker lane was removed to avoid confusion caused by the inappropriate DNA marker.

> *The authors investigated ribozyme cleavage speeds by measuring the observed rate constants under single-turnover conditions. To achieve single-turnover conditions enzyme has to be used in excess over substrate. Usually, the ratios reported in the literature range between 20:1 (from the authors citation list e.g.: for twister sister (Roth et al 2014) and hatchet (Li et al. 2015)) or even ~100:1 (for pistol: Harris et al 2015, or others https://www.sciencedirect.com/science/article/pii/S0014579305002061). Can the authors please share their experimental evidence that only 5:1 excess of enzyme over the substrate as used in their experiments truly creates single-turnover conditions?*

We greatly appreciate the Reviewer for raising this question. To address this, we performed kinetic analysis using different enzyme to substrate ratios (Figure S13). There is not too much difference in kobs, except that kobs reach the highest value of 0.048 min-1 when using 100:1 excess of enzyme over the substrate.

Line 458-460: The kobs of LINE-1-core under single-turnover condition was ~0.05 min-1 when measured in 10mM MgCl2 and 100mM KCl at pH 7.5 (Figure S13).

> *Citations:*
>
> *In the introduction citation number 12 (Roth et al 2014) is mentioned with the CPEB3 ribozyme introduction. This is the wrong citation. Please also insert citations for OR4K15 and IGF1R and LINE-1 ribozyme in this sentence.*

We thank the reviewer for pointing this out and we now have fixed it.

> *Also in the introduction, a hammerhead ribozyme in the 3' UTR of Clec2 genes is mentioned and reference 16 (Cervera et al 2014) is given, I think it should be reference 9 (Martick et al 2008)*

We thank the reviewer for pointing this out and we now have fixed it.

> *In the results section it is stated that, "original sequences were generated from a randomly fragmented human genomic DNA selection based biochemical experiment" citing reference 12. This is the wrong reference, as I could not find that Roth et al 2014 describe the use of such a technique. The same sentence occurs in the introduction almost verbatim (see also minor points).*

We thank the reviewer for pointing this out and we now have fixed it.

> *Minor points*
>
> *Headline:*
>
> *Either use caps for all nouns in the headline or write "self-cleaving ribozyme" uncapitalized*

We thank the reviewer for pointing this out and we now have fixed it.

> *Abstract:*
>
> *1st sentence: in "the" human genome*
>
> *"Moreover, the above functional regions are..." - the word "above" could be deleted here*
>
> *"named as lantern for their shape"- it should be "its shape"*
>
> *"in term of sequence and secondary structure"- "in terms"*
>
> *"the nucleotides at the cleavage sites" - use singular, each ribozyme of this class has only one cleavage site*

We thank the reviewer for pointing these out and we now have fixed them.

> *Introduction:*
>
> *Change to "to have dominated early life forms"*
>
> *Change to "found in the human genome"*
>
> *Please write species names in italics (D. melanogaster, B. mori)*
>
> *Please delete "hosting" from "...are in noncoding regions of the hosting genome"*
>
> *Please delete the sentence fragment/or turn it into a meaningful sentence: "Selection-based biochemical experiments (12).*
>
> *Change to "in terms of sequence and secondary structure, suggesting a more"*
>
> *Please reword the last sentence in the introduction to make clear what is referred to by "its", e.g. probably the homology model of lantern ribozyme generated from twister sister ribozymes?*

> *Please refer to the appropriate methods section when explaining the calculation of RA and RA'.*

We thank the reviewer for pointing these out and we now have fixed them.

> *The last sentence of the second paragraph in the second section of the results states that the authors confirmed functional regions for LINE-1 and OR4K15, however, until that point the section only presents data on LINE-1. Therefore, OR4K15 should not be mentioned at the end of this paragraph.*

In response to the reviewer's suggestions, we have removed OR4K15 from this paragraph.

Line 225-228: The consistency between base pairs inferred from deep mutational scanning of the original sequences and that of the identified functional regions confirmed the correct identification of functional regions for LINE-1 ribozyme.

> *Change to "Both ribozymes have two stems (P1, P2), to internal loops ..."*

We thank the reviewer for pointing this out and we now have fixed it.

> *The section naming the "functional regions" of LINE-1 and OR4K15 lantern ribozymes should be moved after the section in which the circular permutation is shown and explained. Therefore, the headline of section three should read "Consensus sequence of LINE-1 and OR4K15 ribozymes" or something along these lines.*

We thank the reviewer for pointing this out and we now have fixed it.

Line 308-309: Given the identical lantern-shaped regions of the LINE-1-rbz and OR4K15-rbz ribozyme, we named them twister sister-like (TS-like) ribozymes.

> *The statement on the difference between C8 in OR4K15 and U38 in LINE-1 should be further classified. As U38 is only 95% conserved. Is it a C in those other instances or do all other nucleotide possibilities occur? Is the high conservation in OR4K15 an "artifact" of the low mutation rate for this RNA in the deep mutational scanning?*

We thank the reviewer for this question. Yes, the high conservation in OR4K15 an "artifact" of the low mutation rate for this RNA in the deep mutational scanning. That is why RA' value is more appropriate to describe the conservation level of each position. We also mentioned this in the manuscript:

Line 287-288: The only mismatch U38C in L1 has the RA' of 0.6, suggesting that the mismatch is not disruptive to the functional structure of the ribozyme.

> *Section five, first paragraph: instead of "two-stranded LINE-1 core" use the term "bimolecular", as it is more commonly used.*

We thank the reviewer for pointing this out and we now have changed it.

> *Figure caption 3 headline states "Homology modelled 3D structure..."but it also shows the secondary structures of LINE1, OR4K15 and twister sister examples.*

We thank the reviewer for pointing this out and we now have removed “3D”.

> *In Figure 3C, we see a nucleobase labeled G37, however in the secondary structure and sequence and 3D structural model there is a C37 at this position. Please correct the labeling.*

We thank the reviewer for pointing this out and we now have fixed it.

> *Section 7 "To address the above question…" please just repeat the question you want to address to avoid any confusion to the reader.*

We thank the reviewer for pointing these out and we have re-phrased this sentence.

Line 364: Considering the high similarity of the internal loops, we further investigated the mutational effects on the internal loop L1s.

> *Please rephrase the sentence "By comparison, mutations of C62 (…) at the cleavage site did not make a major change on the cleavage activity…", e.g. "did not lead to a major change" etc.*
>
> *Section 8, first paragraph: This result further confirms that the RNA cleavage in lantern…", please delete "further"*
>
> *Change to "analogous RNAs that lacked the 2' oxygen atom in the -1 nucleotide"*
>
> *Methods*
>
> *Change to "We counted the number of reads of the cleaved and uncleaved…"*
>
> *Change to "…to produce enough DNA template for in vitro transcription."*
>
> *Change to "The DNA template used for transcription was used…" (delete while)*

We thank the reviewer for pointing these out and we now have fixed them.

> *Supplement*
>
> *All supplementary figures could use more detailed Figure legends. They should be self-explanatory.*
>
> *Fig S1/S2: how is "mutation rate" defined/calculated?*

We thank the reviewer for pointing this out and we now have added a short explanation. The mutation rate was calculated as the proportion of mutations observed at each position for the DNA-seq library.

> *Fig S3/S4: axis label "fraction", fraction of what? How calculated?*

We thank the reviewer for pointing this out and we now have added a short explanation. The Y axis "fraction" represents the ratio of each mutation type observed in all variants.

> *Fig S5: RA and RA' are mentioned in the main text and methods, but should be briefly explained again here, or it should be clearly referred to the methods. Also, the axis label could be read as average RA' divided by average RA. I assume that is not the case. I assume I am looking at RA' values for LINE-1 rbz and RA values for LINE-1-ori? Also, mention that only part of the full LINE-1-ori sequence is shown…*

We thank the reviewer for pointing this out and we have now added a short explanation. The Y axis represents RA’ for LINE-1-rbz, or RA for LINE-1-ori. The part shown is the overlap region between LINE-1-rbz and LINE-1-ori. We apologize for any confusion caused by our previous statement.

> *Fig S9 the magenta for coloring of the scissile phosphate is hard to see and immediately make out.*

We thank the reviewer for pointing this out and we now have added a label to the scissile phosphate.

> *Fig S10: Why do the authors only show one product band here? Instead of both cleavage fragments as in Figure 5?*

We thank the reviewer for this question. We purposely used two fluorophores (5’ 6-FAM, 3’ TAMRA) to show the two product bands in Figure 5. In Fig S10, long-time incubation was used to distinguish catalysis based self-cleavage from RNA degradation. This figure was prepared before the purchasing of the substrate used in Figure 5. The substrate strand used in Fig S10 only have one fluorophore (5’ 6-FAM) modification. And the other product was too short to be visualized by SYBR Gold staining.

> *Fig S13: please indicate meaning of colors in the legend (what is pink, blue, grey etc.)*
>
> *Please change to "RtcB ligase was used to capture the 3' fragment after cleavage...."*

We thank the reviewer for pointing this out and we now have fixed it.

https://doi.org/10.7554/eLife.90254.2.sa0